

Do surface and bed observables determine basal friction?

A point-wise test in the Amundsen Sea sector

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ABSTRACT. Accurate projections of ice-sheet contributions to sea-level rise are hindered by uncertainties in basal-sliding parameterizations. Although inverse methods can calibrate basal friction to reproduce observed velocities, the controls on inversion-derived friction remain poorly understood. We test whether surface, bed, and geophysical observables can define a consistent point-wise relationship with the inversion-derived basal-friction control C across the Amundsen Sea sector of West Antarctica. Using a sector-wide Icepack inversion as the reference, we train multilayer perceptrons (MLPs) in ten experiments, each with a spatially separated 50 km square and surrounding buffer withheld from training and validation. Each predicted C field is tested in an Icepack forward simulation against observed velocity. The two ice-product configurations improve on a spatially uniform- C reference in eight of ten squares, while the combined configurations do so in nine of ten. Performance varies among locations, adding predictors does not consistently improve performance, and failures occur even where marginal and joint predictor support is high. The tested local predictors produce friction controls that improve modeled velocity in most withheld squares, but the learned relationships are not consistently successful across locations. These experiments highlight the limitations of constructing point-wise basal-friction parameterizations from these observables alone.

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INTRODUCTION

Ice-sheet models are used to project sea-level rise, reconstruct past ice-sheet behavior, and investigate the physical processes governing ice-mass change, e.g. Lecavalier and Tarasov (2025); Nowicki and others (2016). These models contain structural uncertainty arising from incomplete representations of physical processes and parameter uncertainty arising from imperfectly constrained model parameters (Hebeler and others, 2008; Recinos and others, 2023). Both sources of uncertainty limit our ability to constrain probabilistic projections, especially for low-probability, high-impact outcomes (Kopp and others, 2017).

Basal slip is an important source of both structural and parameter uncertainty (Joughin and others, 2019; Bulthuis and others, 2019). Motion at the ice–bed interface can involve sliding over hard bedrock, deformation of subglacial sediment, and changes in basal water pressure, among other processes (Schoof and Hewitt, 2013; Noble and others, 2020). Ice-sheet models generally represent these processes through a sliding law containing one or more spatially varying friction parameters (Weertman, 1964; Schoof, 2005). In an inversion, these parameters are adjusted to reduce the mismatch between modeled and observed quantities such as velocity (MacAyeal, 1993). The resulting friction field can therefore absorb unresolved basal processes, including bed roughness, till rheology, and hydrology (Tulaczyk and others, 2000; Lu and Kingslake, 2024), as well as errors in geometry and differences among model formulations (Barnes and others, 2021). It is consequently an effective model parameter rather than a direct observation of basal conditions.

Inversion studies show that basal resistance across Antarctica is spatially heterogeneous (Morlighem and others, 2013; Zhao and others, 2018). This motivates a working hypothesis: at the resolution of an ice-sheet model, the dimensionless basal-friction control $C(\mathbf{x})$ may be approximated by a point-wise function of observables evaluated at the same location,

$$C(\mathbf{x}) \approx f(\mathbf{z}(\mathbf{x})), \quad (1)$$

where $\mathbf{z}(\mathbf{x})$ contains locally measured or inferred quantities that may influence basal resistance or serve as proxies for unresolved basal conditions. Equation 1 is a testable point-wise approximation at the model-grid scale, not a physical sliding law. We test it by fitting the relationship outside a held-out region and asking whether the predicted C improves modeled velocity within that region, particularly where similar predictor values are well represented in the training data.

We divide the candidate predictors into two groups. The ice-product group describes ice geometry, its gradients, driving stress, and surface air temperature. The geophysical-product group describes bed geometry, bed-surface gradient alignment, geothermal heat flux, and gravity and magnetic anomalies. Ice thickness controls overburden pressure and normal stress at the bed, while surface slope governs driving stress; together, variations in thickness and surface elevation gradient impose systematic constraints on basal drag (MacAyeal, 1989; Cuffey and Paterson, 2010, Sections 7.2–7.4 and 8.2). Surface air temperature provides a large-scale upper thermal boundary condition that influences englacial and basal thermal state over long timescales. Where basal ice reaches the pressure-melting point, basal water pressure can reduce effective pressure and frictional resistance (Cuffey and Paterson, 2010, Chapters 6–7; Iken, 1981; Dawson and others, 2022).

Geophysical factors can affect basal drag through bed topography and substrate composition. Bed gradients characterize topographic variation that can contribute to form drag, while deformable till can reduce resistance and enable sliding (Cuffey and Paterson, 2010; Tulaczyk and others, 2000; Boulton and others, 2001; Hoffman and others, 2022). Inferred friction can also depend on assumptions about ice rheology and subglacial hydrology (Rathmann and Lilien, 2022; McArthur and others, 2023). Gravity disturbances reflect subsurface density variations associated with bedrock and sedimentary structure (Hackney and Featherstone, 2003). Gravity data can therefore constrain broad bed geometry where direct measurements are sparse (Boghosian and others, 2015; Kennedy, 2020). Bedrock lithology can influence sediment characteristics, while deformable sediment can reduce basal resistance (Crompton and Flowers, 2016; Boulton and others, 2001). Gravity measurements cannot resolve the thin (centimeter to meter scale) sediment layers that affect sliding, but larger density contrasts can identify sedimentary basins; such sediment bodies can influence ice-stream onset (Kennedy, 2020; Peters and others, 2006). Magnetic anomalies reflect contrasts in crustal magnetization (Golynsky and others, 2018). In concert with gravity and radar data, they can help infer subglacial geology (Jordan and others, 2023) and regional thermal structure (Martos and others, 2017; Reading and others, 2022), thereby providing indirect context for basal drag. Geothermal heat flux supplies heat at the bed and can influence whether basal ice reaches the pressure-melting point. Where basal ice is at the melting point, additional heat can support basal meltwater production and thereby influence basal resistance (Dawson and others, 2022; Kang and others, 2022).

Previous studies have compared inferred basal resistance with radar, seismic, and other observations of the subglacial environment, but have not established a generally predictive local relationship. Haris and

80 others (2024) found differences in basal shear stress among regions with distinct radar signatures but did
 81 not identify a direct predictive relationship. Das and others (2023) found no robust correlation between
 82 inferred sliding parameters and radar-derived basal reflectivity over Thwaites Glacier, while Kyrke-Smith
 83 and others (2017) found only weak kilometer-scale association between basal drag and seismic acoustic
 84 impedance along Pine Island Glacier (PIG).

85 These studies, together with our understanding of basal processes, suggest that the relationship be-
 86 tween basal conditions and basal resistance may be complex and nonlinear, involving thermo-mechanical
 87 feedbacks, subglacial hydrology, and spatially variable bed properties. Following the field inversion and
 88 machine learning approach of Parish and Duraisamy (2016), we use machine learning to fit the proposed
 89 nonlinear relationship between the selected observables and the inversion-derived friction control C .

90 We use a multilayer perceptron (MLP) as a flexible nonlinear representation of f in Eqn 1. The
 91 point-wise architecture asks whether the values of the selected observables at a location contain enough
 92 information to determine the inversion control there without prescribing a particular functional form.
 93 Observed velocity is deliberately excluded from the predictor set. Because changing C changes the modeled
 94 velocity, including velocity as an input would introduce a self-consistency problem and would make the
 95 inferred relationship depend directly on the velocity used to evaluate it.

96 Recent machine-learning studies in glaciology have addressed related but distinct inverse and constitu-
 97 tive problems. Juvet (2023) inverted a convolutional emulator of Stokes flow, while Juvet and Cordonnier
 98 (2023) and Cheng and others (2024) incorporated ice-flow physics into learned forward or inverse models.
 99 Wang and others (2025) used remote sensing and physics-informed deep learning to infer spatially varying
 100 ice-shelf rheology. These approaches use spatial structure or governing equations to reproduce ice-flow
 101 fields or solve inverse problems. Here, machine learning is used only to fit the local relationship in Eqn 1.

102 We evaluate that hypothesis in ten spatially separated 50 km squares across the Amundsen Sea sector.
 103 Each square and its surrounding buffer are excluded from MLP training and validation. The predicted C
 104 field is then used in a forward ice-flow simulation, and the resulting velocity is compared with observed
 105 velocity. All experiments use the same sector-wide reference inversion, constructed using observed velocity
 106 from the complete sector. The square experiments therefore test whether a point-wise relationship fitted
 107 elsewhere in the sector remains useful in a withheld location. We compare six predictor configurations and
 108 quantify how well the withheld predictor values are represented in the training data. We also withhold
 109 all of Pine Island Glacier in a secondary catchment-scale test. Because inversion-derived C is not neces-

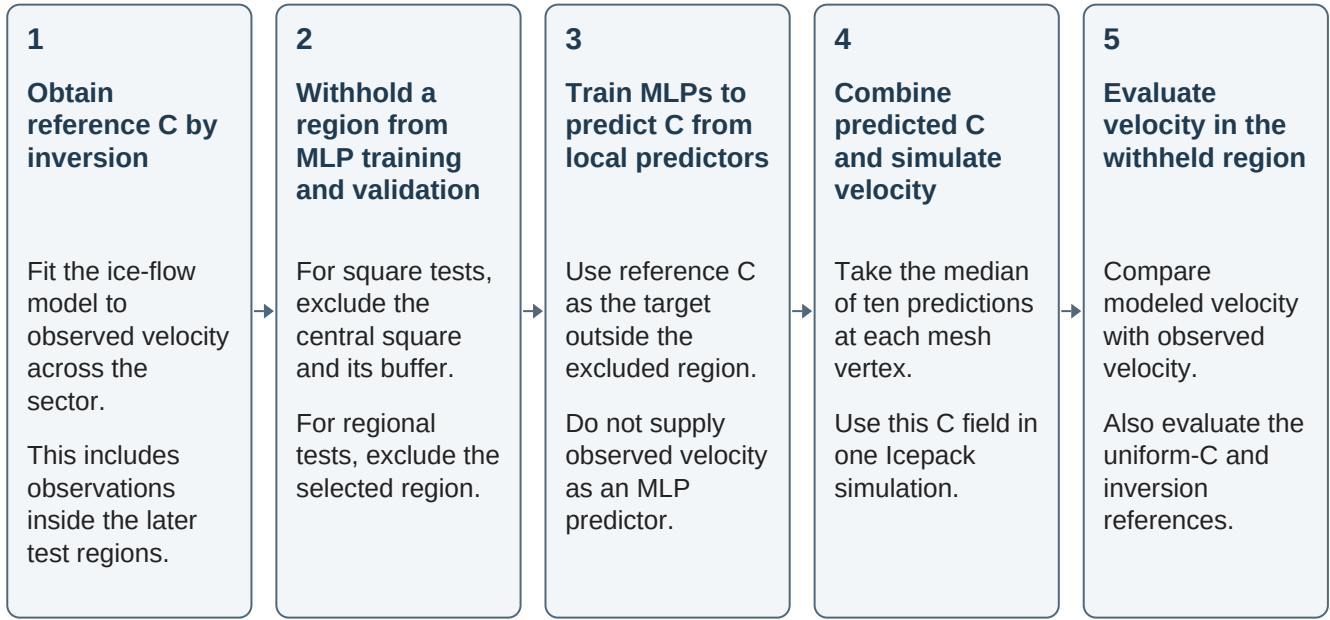


Fig. 1. Steps in the spatial-transfer experiment. Each experiment–configuration pair uses one modeled velocity field obtained from the vertex-wise median of ten predicted basal-friction controls. Predicted C replaces reference C only within the withheld footprint or catchment, intersected with the eligible grounded area; reference C is retained elsewhere. Observed velocity enters the inversion and the final velocity comparison, but is not an MLP predictor.

sarily unique (MacAyeal, 1993), modeled velocity provides the primary evaluation, while direct agreement with the reference C is treated as secondary (Appendix section Basal-friction control: distributions and comparisons).

METHODS

The experiment links a sector-wide basal-friction inversion with spatially withheld tests of the learned relationships (Fig. 1). The inversion provides the reference C used to train the MLPs outside each excluded region. Predicted C is then supplied to Icepack, and the resulting modeled velocity is compared with observations in the withheld region.

Study region and observations

The study domain covers the Pine Island, Thwaites, and Dotson–Crosson drainage systems in the Amundsen Sea sector of West Antarctica (Fig. 2). The catchment outlines were delineated manually, with lateral boundaries following glacier velocity flowlines. PIG, Thwaites, and Dotson are geographic labels; they do not imply that each catchment contains one basal regime. Two non-overlapping inter-catchment corridors complete the five-region division used for regional comparisons.

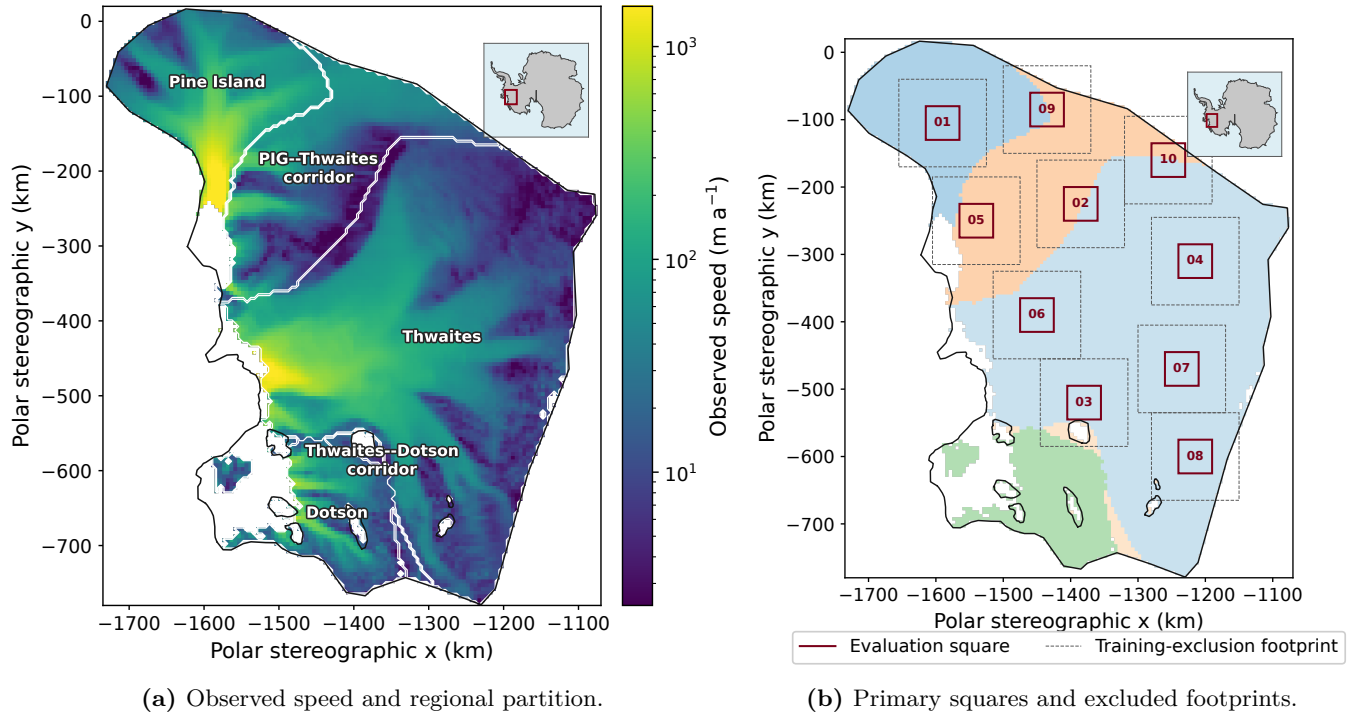


Fig. 2. Study region and experimental geometry. Panel (a) shows observed MEaSUREs speed over the eligible part of the Amundsen Sea sector, with the PIG, Thwaites, and Dotson regions and the two inter-catchment corridors. Panel (b) shows the ten 50 km primary evaluation squares (solid) and their 130 km excluded footprints (dashed). Each footprint includes a 40 km buffer on every side of the central square and is excluded from MLP training and validation.

Velocity is taken from the Making Earth System Data Records for Use in Research Environments (MEa-
 SUREs) Phase-Based Antarctica Ice Velocity Map, derived from satellite radar interferometry (Mouginot
 and others, 2019). Surface elevation, ice thickness and bed elevation are taken from BedMachine Antarc-
 tica, version 2 (Morlighem, 2020; Morlighem and others, 2020). BedMachine combines direct observations
 with reconstruction that uses ice velocity in some regions. The ice-product predictors therefore represent
 available geometry products, some of which already incorporate velocity information, although velocity
 itself is not used as a predictor. BedMachine uses different methods to reconstruct ice thickness be-
 tween observations. Most of our test regions (detailed in following sections) lie in regions mapped using
 streamline-diffusion interpolation, which favors smoother thickness variations along flow than across flow
 (Morlighem and others, 2020).

Surface air temperature and geothermal heat flux follow the products distributed with Quantarctica
 (Matsuoka and others, 2018; Comiso, 2000; An and others, 2015). Magnetic anomaly is taken from the
 5 km EPSG:3031 ADMAP-2S grid (Eagles and others, 2024), and gravity disturbance from the oriented
 EPSG:3031 AntGG2021 gravity-anomaly grid (Scheinert and others, 2024). Interpolation to the model grid
 does not add spatial information beyond that contained in the original products, whose native resolutions
 and processing differ.

Ice-flow model and basal-friction inversion

All simulations were performed with Icepack (Shapero and others, 2021), using a depth-integrated stress
 balance based on the Shallow Shelf Approximation (SSA) (MacAyeal, 1989). The shallow-shelf approxi-
 mation neglects vertical shear, so the modeled horizontal velocity is compared directly with the observed
 surface velocity. Because the inferred basal-friction control depends on the adopted stress-balance approx-
 imation and basal law, the model parameter C_b is model-dependent rather than a unique representation
 of basal conditions.

Basal friction is represented with a Weertman-type law,

$$\boldsymbol{\tau}_b = -C_b \|\mathbf{u}\|^{1/n-1} \mathbf{u}, \quad (2)$$

where $\boldsymbol{\tau}_b$ is the resisting basal shear stress on the ice, $\mathbf{u}$ is depth-averaged horizontal velocity, $n = 3$ is the ex-
 ponent used in the basal law, and C_b is a dimensional basal-friction coefficient with units $\text{MPa (m a}^{-1}\text{)}^{-1/3}$.
 A Weertman law is a simplified hard-bed parameterization and does not explicitly represent processes such

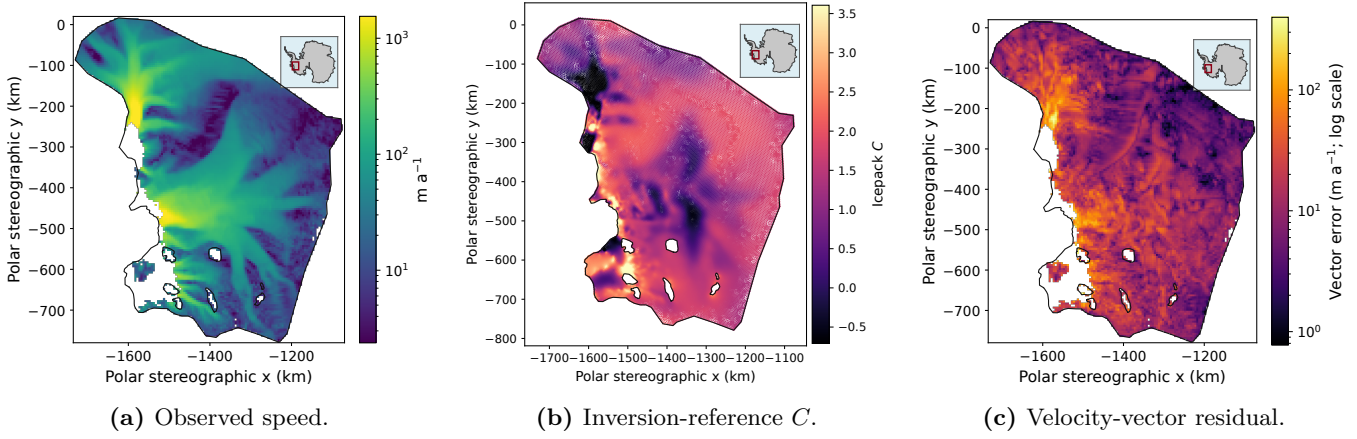


Fig. 3. Sector-wide inversion used to construct the learning target. The inversion field is one regularized solution for the stated data, model equations, sliding law, boundary conditions, and regularization, and provides the reference C for the learning experiments. The black line is the model-domain boundary.

as effective-pressure dependence, cavitation, or deforming till. We hold the basal law fixed so that the experiment tests the proposed mapping with one consistent ice-flow model.

Observed velocities are prescribed along the non-terminus boundaries, while an ocean-pressure traction condition is applied at the termini. Positivity and the grounding transition are imposed through

$$C_b = \phi C_0 e^C, \quad (3)$$

where C is the dimensionless inversion control used to represent spatial variation in basal friction and $C_0 = 0.01 \text{ MPa (m a}^{-1})^{-1/3}$. ϕ is a height-above-flotation ramp that continuously reduces basal resistance as flotation is approached and enforces $C_b = 0$ over floating ice (Leguy and others, 2014; Seroussi and others, 2014). The full stress balance and grounding transition are described in Appendix section Ice-flow equations, grounding transition, and inversion method.

The inversion minimizes the total objective function

$$J(\mathbf{u}, C; r_C) = \frac{1}{2N} \sum_{i=1}^N \|\mathbf{u}(\mathbf{x}_i) - \mathbf{u}_{\text{obs}}(\mathbf{x}_i)\|^2 + \frac{1}{2|\Omega|} \left(\frac{L}{r_C} \right)^2 \int_{\Omega} \|\nabla C\|^2 dA, \quad (4)$$

where N is the number of paired-velocity observations, $|\Omega|$ is the domain area, $L = 7.5 \text{ km}$ a fixed reference length, and r_C controls smoothness. Smaller r_C gives stronger smoothing. Following the use of L-curves in ice-sheet inverse models (Wolovick and others, 2023), we selected $r_C \simeq 0.014$ using a maximum-curvature criterion (Appendix Fig. 8). Gradients of Eqn 4 are obtained with the adjoint method. The resulting field (Fig. 3) is used for every learning experiment.

Learning dataset and predictor configurations

We construct a common learning dataset by collocating the mesh-interpolated predictors with the inversion-reference C on the observation grid. Each row contains the predictors and reference C at one observation-grid location. Eligible rows are required to have finite values for all twelve predictors and both velocity components, with $\phi > 0.1$ to exclude floating ice and the immediate grounding transition. The same eligibility criteria are used for all predictor configurations and experiments. Observed velocity is not included as a predictor; the twelve predictors and their six configurations (CFG01–CFG06) are summarized in Table 1. Gradient construction and exact row counts are given in Appendix section Eligible rows, predictor construction, and support.

The additional predictor in CFG06 measures bed–surface gradient alignment,

$$a_{bs} = \frac{\nabla b \cdot \nabla s}{\sqrt{(\|\nabla b\|^2 + \epsilon_b^2)(\|\nabla s\|^2 + \epsilon_s^2)}}, \quad (5)$$

where $\epsilon_b = 5.4 \times 10^{-3}$ and $\epsilon_s = 1.5 \times 10^{-4} \text{ m m}^{-1}$. When both gradients are large relative to these values, a_{bs} approaches $\cos \theta_{bs}$. The added terms prevent the alignment from becoming unstable where either gradient is very small. This predictor describes relative alignment without using velocity direction. The choice of ϵ_b and ϵ_s is described in Appendix section Eligible rows, predictor construction, and support.

CFG01 contains all ice-product predictors and CFG02 the selected ice-product subset. CFG03 contains all geophysical-product predictors and CFG04 the selected geophysical-product subset. CFG05 combines the two selected subsets, and CFG06 adds Eqn 5. The subsets used in CFG02 and CFG04 were selected from preliminary catchment-scale analyses of modeled velocity conducted before the present spatially withheld experiments and were not modified based on the square results.

Spatially withheld square experiments and predictor support

The primary experiment uses ten 50 km squares distributed across the study region (Fig. 2b). For each square, the complete 130 km footprint is excluded from MLP training and validation, leaving a 40 km buffer on every side of the central evaluation square. The ten excluded footprints do not overlap. Following the rationale of spatially separated evaluation (Roberts and others, 2017; Valavi and others, 2019), the buffer reduces dependence on nearby spatially correlated training rows. It does not remove the dependence introduced by the common sector-wide inversion or by non-local ice flow. The central 50 km square is

Table 1. Predictors in the six configurations. Ice-product predictors are denoted I and geophysical-product predictors G. A filled circle indicates inclusion.

Symbol	Predictor	Unit	Group	CFG01	CFG02	CFG03	CFG04	CFG05	CFG06
s	surface elevation	m	I	•	•			•	•
h	ice thickness	m	I	•	•			•	•
$\ \nabla s\ $	surface-gradient magnitude	1	I	•	•			•	•
$\ \nabla h\ $	thickness-gradient magnitude	1	I	•	•			•	•
τ_d	driving stress	MPa	I	•					
T_s	surface air temperature	°C	I	•	•			•	•
b	bed elevation	m	G			•	•	•	•
$\ \nabla b\ $	bed-gradient magnitude	1	G			•	•	•	•
Q_b	geothermal heat flux	mW m ⁻²	G			•	•	•	•
g_d	gravity disturbance	mGal	G			•			
m_a	magnetic anomaly	nT	G			•			
a_{bs}	bed-surface gradient alignment	1	G						•

the primary evaluation area; the surrounding buffer and complete footprint are used only for maps and secondary summaries, not as additional independent tests. The quantitative basis for the buffer is given in Appendix section Spatial separation and buffer choice.

Candidate square locations were selected using spatial separation and predictor-support criteria so that the withheld predictor values were generally well represented in the remaining training region. Support was assessed using both marginal and joint criteria. No criterion based on the inversion-reference C or modeled velocity errors was used in square selection. The distribution of inversion-reference C was examined only afterward because the reference field is one regularized, non-unique inversion solution, and a different predicted C may still reproduce the observed velocity. Modeled velocity errors were excluded to avoid selecting held-out regions based on how well the model ultimately performed.

Marginal support requires every included predictor at a held-out point to lie within the 1st–99th percentile range of the corresponding eligible non-held-out rows. Joint support asks whether the complete predictor combination has a sufficiently close analogue in the remaining training region. We report the area fraction satisfying both criteria and retain every held-out point in the modeled velocity evaluation regardless of support. Details of the joint-support transformation and distance threshold are given in Appendix section Eligible rows, predictor construction, and support. A separate regional experiment

withholds the two inter-catchment corridors and is reported in Appendix section Basal-friction control: distributions and comparisons.

Catchment-scale transfer test

In addition to the ten compact spatial tests, we retained Pine Island Glacier (PIG) as one catchment-scale transfer test chosen before its results were examined. This tests whether a point-wise relationship learned elsewhere in the Amundsen Sea sector can be transferred across an entire glacier catchment. All eligible PIG rows were excluded from MLP training and validation, while the remaining grounded sector, including Thwaites, Dotson–Crosson, and the two inter-catchment corridors, provided the training and validation rows. Predictor support in PIG was evaluated using the same marginal and joint criteria as the square experiments and without using the inversion control or velocity errors.

Model fitting and repeated realizations

A point-wise regression model provides a direct way to test whether the inversion-derived friction control can be expressed as a local function of the available predictors, without introducing additional spatial information from neighboring locations. We use a MLP as a flexible nonlinear representation of $f(\cdot)$ in Eqn 1. More spatially structured architectures, such as convolutional or graph neural networks, could use information from neighboring locations, but would test a different hypothesis from the local relationship considered here.

The MLP contains ten hidden layers of 200 units and predicts the inversion-reference C from the selected predictor configuration. Inputs and targets are robustly scaled using medians and interquartile ranges calculated from the training rows only. The network architecture, optimization settings, model-selection procedure, and training and validation histories are given in Appendix section MLP training and statistical comparisons.

For each square experiment, the eligible rows remaining after removal of the excluded footprint are divided reproducibly into 90% for training and 10% for validation. Nearby training and validation rows are not statistically independent because of interpolation, predictor resolution, spatial autocorrelation, and regularization of the inversion-reference field. The random split is therefore used only for model fitting and selection, while performance is evaluated separately in the spatially withheld central square. Exact row counts are given in Appendix section Eligible rows, predictor construction, and support.

For each experiment and predictor configuration, we train ten reproducible realizations that differ in the training/validation split, initial weights, and epoch shuffling. Using multiple realizations reduces sensitivity to any one set of training choices and provides a more stable estimate of the predicted C field. Matching realizations use the same row split across predictor configurations to support direct comparison. The spread among realizations reflects sensitivity to these training choices rather than calibrated predictive uncertainty or independent spatial variability. The primary prediction is the vertex-wise median of the ten predicted C fields, which is then evaluated in a separate Icepack forward simulation.

Forward simulations and measures of performance

For each experiment and predictor configuration, the vertex-wise median of the ten predicted C fields is evaluated in a separate Icepack forward simulation. For each square, predicted C replaces the inversion-reference C only within the complete 130 km excluded footprint, intersected with the common eligible grounded mask. The inversion-reference C is retained everywhere outside that footprint. The corresponding uniform- C simulation uses the identical replacement mask, so the two simulations differ only in the spatial pattern of C inside the excluded footprint. The PIG experiment uses the same procedure over the withheld catchment. This controlled evaluation comprises 60 square simulations using predicted C , one PIG simulation using predicted C , ten square-specific uniform- C simulations, and one PIG uniform- C simulation.

The primary measure of performance is vector velocity RMSE (hereafter velocity RMSE) against the original paired MEaSUREs velocity components at their regular 450 m pixel centres over the evaluation area A . Modeled velocity is evaluated at the same locations. This evaluation is distinct from the mesh-interpolated observation field used by the solver for inversion and boundary conditions. The spatial average below is approximated using the mean squared vector residual at these equally spaced points (Appendix section MLP training and statistical comparisons):

$$\text{RMSE}_u = \left[\frac{\int_A \|\mathbf{u}_{\text{ML}} - \mathbf{u}_{\text{obs}}\|^2 dA}{\int_A dA} \right]^{1/2}. \quad (6)$$

For the maps, we also use the local vector-error magnitude

$$e(\mathbf{x}) = \|\mathbf{u}(\mathbf{x}) - \mathbf{u}_{\text{obs}}(\mathbf{x})\|.$$

The spatially uniform- C reference provides a predictor-free baseline analogous to a mean-only model. It uses a single control value equal to the finite-element-area-weighted mean inversion-reference C over the eligible non-held-out region. For squared error in C , this mean is the best-fitting constant over that region, while containing no predictor-dependent spatial variation. Because $C_b \propto \exp(C)$, averaging C corresponds to a geometric mean of $C_0 \exp(C)$ before multiplication by the grounding ramp ϕ . The constant is not fitted to held-out velocity and is applied over the same withheld replacement region as the predicted field, while the inversion-reference C is retained outside that region in both simulations. The comparison therefore asks whether spatial variation in C inferred from the predictors improves modeled velocity beyond what is obtained from a single representative friction control. We also report relative RMSE, $\text{RMSE}_{\text{ML}}/\text{RMSE}_{\text{uniform}}$; values below one indicate improvement over the uniform- C reference.

The ten held-out squares are the independent spatial tests. For each predictor configuration, performance is summarized by the median and interquartile range of the ten square RMSE values. Related configurations are also compared square by square to determine whether adding or removing predictors gives a consistent improvement across locations.

RESULTS AND DISCUSSION

Modeled velocity performance

The local observables often produce a C that improves modeled velocity relative to the spatially uniform- C reference, but performance varies substantially among locations. Figure 4 shows the velocity RMSE for each held-out square, including the inversion and uniform- C references. The two ice-product configurations have the lowest median RMSE values across the squares, 32.9 m a⁻¹ for CFG01 and 32.5 m a⁻¹ for CFG02, and each improves on uniform C in eight of ten squares (Table 2). CFG05 and CFG06, which combine the selected ice-product and geophysical-product predictors, improve on uniform C in nine squares, while CFG03 and CFG04, the all- and selected-geophysical configurations, improve on uniform C in six and seven squares, respectively.

The inversion-reference solution has a substantially lower median RMSE of 13.2 m a⁻¹. This is not a withheld benchmark, because the inversion was fitted using velocity observations from the complete sector, including the evaluation squares. It nevertheless provides context for the velocity fit obtained with the spatially resolved inversion. The substantially larger errors from the predicted C fields show that the point-wise mappings recover only part of the improvement available from the inversion-derived spatial

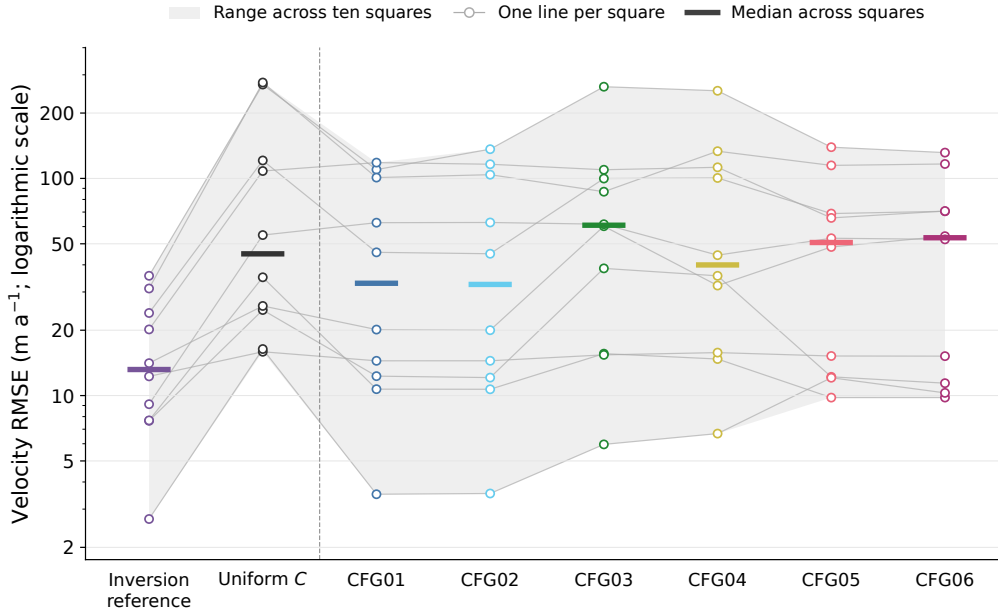


Fig. 4. Velocity RMSE against observed MEaSUREs velocity in the ten central squares. The inversion reference and square-specific uniform- C reference are followed by forward simulations using the median predicted C for all six configurations. Each gray line connects the same square; shading spans the minimum to maximum across squares, and short horizontal marks show the medians. The vertical axis is logarithmic. Uniform C is calculated separately from the eligible region outside each excluded footprint; both the predicted and uniform controls replace the inversion reference only inside that footprint. The inversion reference was fitted using sector-wide observations, including those in the test squares.

structure. At the same time, their frequent improvement over uniform C shows that the predictors contain information relevant to the effective friction control.

Removing driving stress from CFG01 to form CFG02 changes the median velocity RMSE only slightly, from 32.9 to 32.5 m a^{-1} , and both configurations improve on uniform C in eight squares. Because driving stress is derived from ice thickness and surface slope, which are already included in CFG02, supplying it explicitly provides no consistent improvement across the ten squares. Among the geophysical configurations, CFG04 has a lower median velocity RMSE than CFG03 (40.0 versus 60.9 m a^{-1}), but CFG04 has lower velocity RMSE than CFG03 in only five of the ten squares; CFG03 performs better in the other five. Relative to uniform C , CFG03 reduces velocity RMSE in six squares and CFG04 in seven.

Combining the selected predictor groups does not produce a consistent advantage over either group alone. CFG05 performs better than CFG02 in three squares and better than CFG04 in seven. Adding bed-surface gradient alignment in CFG06 again divides the comparison evenly, with CFG06 performing better than CFG05 in five squares and worse in five. The paired sign tests provide no statistical evidence of a consistent advantage in these five configuration contrasts (Appendix section MLP training and statistical comparisons); this does not establish equivalence or negate the improvements over uniform C . Overall,

Table 2. Velocity RMSE against observed MEaSURES velocity in the ten central squares. Values are the median and interquartile range across squares. Relative RMSE uses the corresponding square-specific uniform- C reference. The inversion reference was fitted using sector-wide observations, including those inside the test squares, and therefore represents the fitted reference rather than a withheld prediction. RMSE is in m a^{-1} .

Control/config.	Predictors	Median RMSE	IQR	Better than uniform	Median relative RMSE	Relative RMSE IQR
Uniform C	spatially uniform	44.9	25.0–117.8	–	1.00	1.00–1.00
Inversion reference	–	13.2	8.0–23.0	10/10	0.19	0.17–0.29
CFG01	all ice	32.9	12.8–91.3	8/10	0.45	0.37–0.88
CFG02	selected ice	32.5	12.7–93.7	8/10	0.50	0.37–0.87
CFG03	all geophysical	60.9	21.3–96.6	6/10	0.97	0.54–1.09
CFG04	selected geophysical	40.0	19.9–109.5	7/10	0.88	0.56–1.03
CFG05	selected combined	50.7	13.0–68.1	9/10	0.59	0.50–0.90
CFG06	combined + alignment	53.3	12.4–70.7	9/10	0.61	0.47–0.88

the relative performance of the predictor configurations changes among squares, indicating that no tested predictor set provides a uniformly successful point-wise relationship across the study region.

Predictor support and modeled velocity performance

We next examine whether poor modeled velocity performance is associated with low predictor support. Across the ten primary squares, predictor support is generally high (Fig. 5a). Marginal support covers more than 94% of each primary square, while lower joint support is concentrated mainly in SQ08. High support, however, does not guarantee good modeled velocity performance. Figure 5b shows performance relative to uniform C ; a relative RMSE greater than one means that the predicted control performs worse than the uniform- C reference. SQ07–CFG03, for example, has 99.4% support by both criteria but a relative RMSE of 2.33, while SQ02–CFG04 has 97.3% support and a relative RMSE of 1.44 (Table 3). These failures cannot be explained solely by individual predictors falling outside their training ranges or by the absence of a nearby multivariable match under the adopted definition.

Predictor support indicates whether similar predictor combinations are represented in the training region, but does not distinguish between densely and sparsely sampled parts of predictor space. We therefore examined whether training representation was associated with errors in the predicted C for CFG02 and CFG04 (Appendix section Training representation and prediction error). For CFG02, C RMSE increased with sparser representation in seven of the ten squares, but the overall relationship was weak. The median within-square Spearman correlation between training-representation percentile and absolute

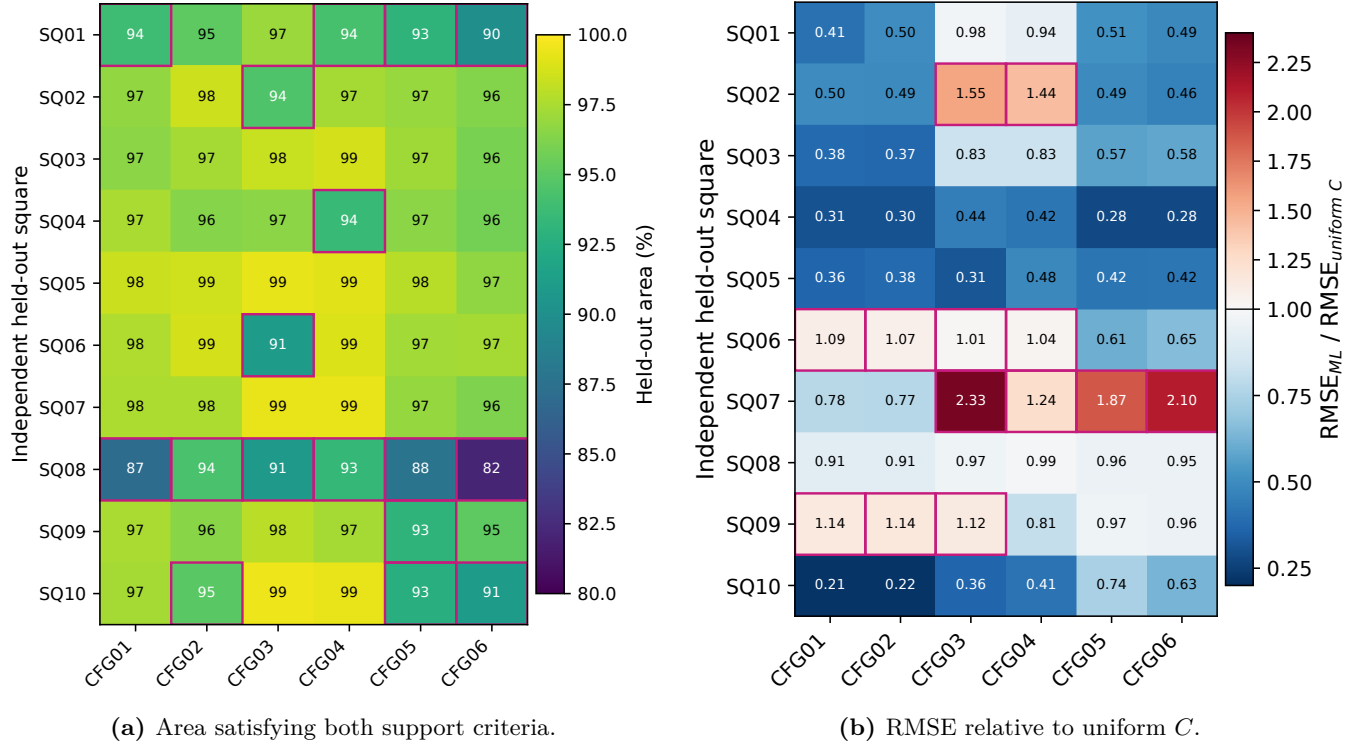


Fig. 5. Predictor support and modeled velocity performance in the ten spatial tests for six predictor configurations. Panel (a) gives the percentage of each held-out area satisfying both marginal and joint support; magenta outlines mark values below the 95% criterion. Panel (b) gives relative RMSE; magenta outlines mark cases that do not improve on the spatially uniform- C reference. Marginal bounds use eligible non-held-out rows, while joint matches use the 5 km reference grid within the training region.

Table 3. Examples of poor performance, defined here as velocity RMSE exceeding the uniform- C reference, despite high predictor support. Support is the percentage of held-out points that satisfy both the marginal and joint criteria.

Square	Config.	Support (%)	ML RMSE	Uniform RMSE	Relative RMSE
(m a ⁻¹)					
SQ02	CFG04	97.3	35.6	24.8	1.44
SQ06	CFG01	97.6	118.3	108.2	1.09
SQ07	CFG03	99.4	60.2	25.9	2.33
SQ09	CFG02	96.4	62.7	54.8	1.14

C error was only 0.069, and the median C RMSE differed by only 0.004 across the three representation categories. CFG04 showed no similar pattern, with a median correlation between representation percentile and absolute C error of -0.052 . Sparse training representation is therefore associated with larger C RMSE in some CFG02 experiments, but the relationship is weak and is not shared by CFG04.

Training representation also does not show a consistent correspondence with modeled velocity performance, as illustrated by the square-level examples in Appendix section Training representation and prediction error. In some squares, modeled velocity has lower RMSE than uniform C in sparsely represented categories but not in better-represented categories. The same Appendix examples also show that closer agreement with the inversion-reference C does not consistently correspond to lower relative RMSE. Training representation therefore does not by itself explain the failures to improve modeled velocity relative to uniform C .

Spatial patterns of success and failure

Figure 6 compares CFG02, the selected ice-product configuration, with CFG04, the selected geophysical-product configuration. The two ice-product configurations have nearly identical summary performance, so CFG02 is used to represent that predictor group without repeating a visually similar map. Each square comes from a separate set of MLPs for which the complete 130 km footprint was excluded from training and validation; the panels therefore assemble ten separately withheld predictions rather than one sector-wide prediction. In every simulation, the predicted and uniform controls replace the inversion reference only inside the corresponding footprint. Across the central 50 km squares, CFG02 has lower velocity RMSE than CFG04 in eight of ten tests. Over the complete 130 km footprints, both configurations improve on uniform C in nine of ten tests, with median relative RMSEs of 0.64 for CFG02 and 0.86 for CFG04. Across the ten central 50 km squares, velocity error exceeds 100 m a^{-1} over 13.0% for CFG02 and 22.8% for CFG04, compared with only 0.13% for the inversion-reference solution. Thus, although the predicted controls often improve on uniform C , regions of large absolute velocity error remain much more extensive than for the fitted inversion-reference control.

The maps show that the advantage of the ice-product configuration occurs across several distinct flow settings. In the fast PIG test SQ01, the central-square RMSE ratio is 0.50 for CFG02 and 0.94 for CFG04, with the largest CFG02 improvement concentrated in the fast-flowing part of the square. The local error difference adds an important distinction: CFG02 reduces error over 84.8% of the central square, with a

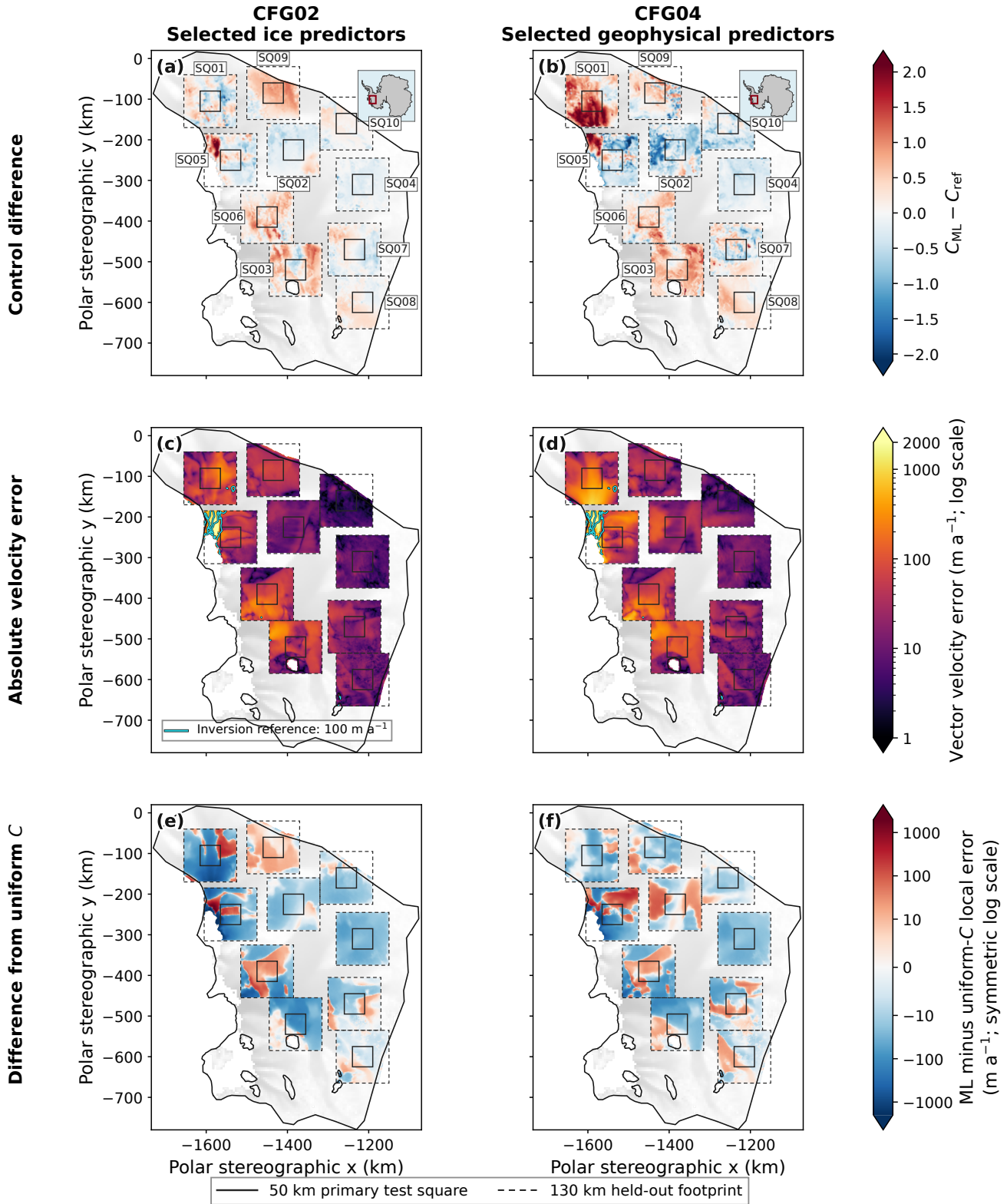


Fig. 6. Spatial comparison of the selected ice-product configuration (CFG02; left) and selected geophysical-product configuration (CFG04; right). Panels (a, b) show $C_{ML} - C_{ref}$. Panels (c, d) show the vector velocity error relative to MEaSUREs observations, $e_{ML} = \|\mathbf{u}_{ML} - \mathbf{u}_{obs}\|$. Panels (e, f) show $e_{ML} - e_{uniform}$; negative values indicate lower local error for the predicted control and positive values indicate lower error for uniform C . Cyan contours in panels (c, d) mark an inversion-reference velocity error of $100 m a^{-1}$. In each experiment, both controls replace the inversion reference only inside the complete 130 km footprint. Velocity errors use the original paired MEaSUREs components and are shown as $1.8 km$ bin means for legibility; all statistics were calculated from the evaluated observation-grid points.

median pointwise reduction of 133.5 m a^{-1} , whereas CFG04 reduces error at every evaluated point but has a much smaller median pointwise reduction of 14.8 m a^{-1} . Thus, the fraction of the square with lower local error does not by itself determine the RMSE improvement; the magnitude and spatial distribution of the error reductions are also important. CFG02 also gives the larger improvement in moderately fast Thwaites flow in SQ03 (0.37 versus 0.83). There, local error is lower than for uniform C over 90.4% of the central square for CFG02 and 77.5% for CFG04. Good performance is not confined to fast flow: both configurations improve substantially on uniform C in the slow Thwaites square SQ04 (0.30 and 0.42) and the slow PIG–Thwaites corridor square SQ10 (0.22 and 0.41).

The large absolute residuals in the SQ05 footprint occur where it extends from the PIG–Thwaites corridor into fast PIG flow. These large residuals contribute disproportionately to the footprint-wide RMSE, yet both learned controls still improve on uniform C (ratios of 0.85 and 0.86). Figure 6 panels (e, f) show that this footprint contains both reductions and increases in local error: CFG02 reduces error over 80.7% of its evaluated area and CFG04 over 60.2%. The large absolute errors in panels (c, d) therefore coexist with improvements relative to the uniform control. The inversion-reference velocity error is also elevated in this part of SQ05, indicating that some of the mismatch remains even with the fitted reference C . Over the complete footprint, however, the velocity RMSEs are 539.5 and 546.2 m a^{-1} for CFG02 and CFG04, compared with 77.1 m a^{-1} for the inversion reference. The predicted controls therefore introduce substantial additional error.

Although CFG02 performs better than CFG04 in most squares, the relative performance of the two predictor groups varies by location. In SQ02, which spans the PIG–Thwaites corridor and Thwaites, CFG02 performs substantially better than the uniform- C reference, whereas CFG04 performs worse. The central-square RMSE ratios are 0.49 for CFG02 and 1.44 for CFG04, and the corresponding full-footprint ratios are 0.58 and 1.91. The local comparison shows that CFG02 reduces error over 99.8% of the central square, whereas CFG04 reduces it over only 42.5%. The higher CFG04 RMSE therefore accompanies a mixture of local improvements and deteriorations, rather than uniformly larger errors. CFG02 also agrees more closely with the inversion-reference control in the central square, with a C RMSE of 0.22 compared with 0.61 for CFG04. SQ09, spanning PIG and the PIG–Thwaites corridor, shows the opposite pattern: CFG04 improves on uniform C while CFG02 does not, with central-square ratios of 0.81 and 1.14 and full-footprint ratios of 0.88 and 1.20, respectively. In this central square, $e_{\text{ML}} - e_{\text{uniform}}$ is positive at every evaluated point for CFG02 and negative at every evaluated point for CFG04. CFG04 also has the lower

central-square C RMSE in SQ09 (0.49 versus 0.74). These examples show that neither predictor group performs better at every location. The central 50 km square remains the primary withheld-region test, while the complete 130 km footprint is used only as a secondary spatial diagnostic.

In SQ06, both configurations are slightly worse than uniform C in the central Thwaites square (ratios of 1.07 and 1.04), despite simultaneous predictor support exceeding 98%. Local error is lower than uniform C over only 12.6% of the central square for CFG02 and 22.6% for CFG04. Over the complete footprint, however, the ratios decrease to 0.62 and 0.79, and local error reductions cover 54.1% and 59.5% of the evaluated area, respectively. The difference maps therefore show how improvements around the central square can accompany poorer performance within it. Similarly, CFG04 changes from a ratio of 1.24 in central SQ07 to 0.80 over the complete footprint. In contrast, the CFG04 failure in SQ02 and the CFG02 failure in SQ09 remain even when the full footprints are evaluated. SQ08 presents a different case: the uniform- C error is already only 15.9 m a^{-1} , compared with 12.3 m a^{-1} for the inversion-reference simulation, leaving relatively little scope for improvement. CFG02 and CFG04 give RMSE values of 14.5 and 15.8 m a^{-1} , respectively. Although local error is lower than for uniform C over 75.0% and 71.3% of the central square, the median local reductions are only 4.3 and 5.0 m a^{-1} . In this case, widespread negative values in panels (e, f) represent small absolute gains, and the larger residuals elsewhere limit the reduction in RMSE.

As discussed above, agreement with the inversion-reference C and velocity performance coincide in SQ02 and SQ09. In fact, when comparing CFG02 with CFG04 within each of the ten central squares, the configuration with lower C RMSE also has lower velocity RMSE. The size of the improvement in velocity RMSE from one configuration to the other, however, does not follow directly from the magnitude of the control error. Over the complete SQ06 footprint, CFG02 and CFG04 have nearly identical C RMSE values (0.435 and 0.439), while their velocity RMSE ratios differ substantially (0.62 and 0.79). Conversely, over the complete SQ05 footprint, the lower C RMSE of CFG02 (0.57 versus 0.85 for CFG04) accompanies only a modest reduction in velocity RMSE (539.5 versus 546.2 m a^{-1}). Closer agreement with the inversion-reference control therefore does not translate into a proportional improvement in velocity RMSE. Within the individual squares, larger absolute local differences from the inversion-reference C generally coincide with larger local velocity errors, although the strength of this relationship varies among locations. The median within-square Spearman correlations are 0.32 for CFG02 and 0.42 for CFG04, indicating a modest positive association rather than a one-to-one correspondence between C error and velocity error.

The spatial pattern of the control difference may help explain this variation. Membrane stresses couple neighboring locations, so a change in C can alter velocity beyond the location where it is applied. Local velocity error can therefore reflect control differences elsewhere as well as those at the same point. The sensitivity of the central-square results to the region over which C is replaced is quantified in Appendix section Sensitivity to the replacement region. Future work could use spatial sensitivity analysis to examine how perturbations to C in different parts of the domain contribute to the velocity response.

Taken together, the ten central-square tests show that the selected ice-product and geophysical-product configurations improve velocity RMSE relative to uniform C in a majority of the squares. Neither configuration improves every square, and some failures persist across the complete withheld footprint despite high predictor support. The selected predictors therefore contain information that can improve modeled velocity relative to a uniform control, but they do not define a point-wise relationship that is consistently successful across locations.

Catchment-scale test at Pine Island Glacier

The secondary catchment-scale experiment tests whether the point-wise relationship remains useful when all of PIG is withheld. We evaluate CFG02, the selected ice-product configuration, using the same ice-flow model and evaluation procedure as in the square experiments. The spatially uniform- C reference produces a vector velocity RMSE of 384.9 m a^{-1} , whereas CFG02 produces an RMSE of 354.8 m a^{-1} . This corresponds to a modest 8% reduction in RMSE relative to the uniform- C reference. The inversion-reference simulation has an RMSE of 48.1 m a^{-1} , showing that the transferred point-wise mapping recovers only a small fraction of the improvement obtained with the spatially resolved inversion.

CFG02 lowers the local velocity error relative to uniform C over 77.6% of the evaluated PIG area (Fig. 7). The improvement occurs across all observed-speed classes. The relative RMSE is 0.59 below 100 m a^{-1} , 0.84 between 100 and 500 m a^{-1} , 0.89 between 500 and 1000 m a^{-1} , and 0.96 above 1000 m a^{-1} . The improvement is therefore spatially widespread, but its influence on the catchment-wide statistic is concentrated in fast flow. The fastest 10% of the evaluated area, corresponding to observed speeds above approximately 504 m a^{-1} , contributes 59.8% of the total squared-error reduction relative to uniform C , but still accounts for 89.5% of the CFG02 squared error. In the fastest flow ($> 1000 \text{ m a}^{-1}$), the RMSE remains 1390.0 m a^{-1} , compared with 1448.8 m a^{-1} for uniform C . Thus, CFG02 improves much of the slow and intermediate-speed catchment interior and parts of the fast-flowing trunk, but provides only a modest

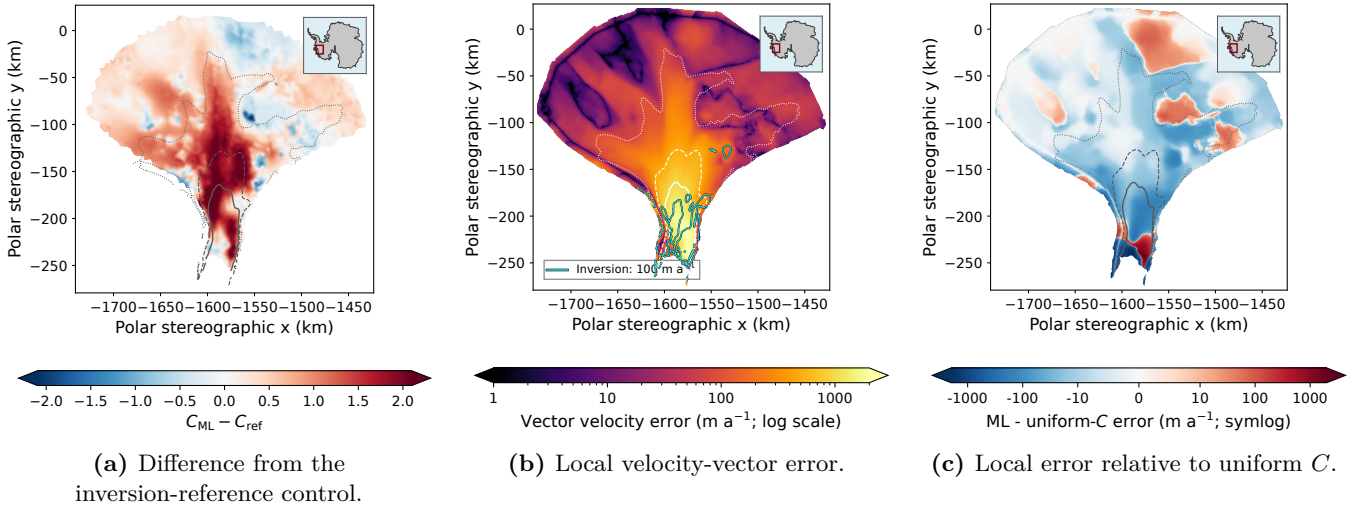


Fig. 7. Spatial evaluation of CFG02 with all of PIG withheld. Panel (a) shows the median predicted control relative to the inversion-reference control. Panel (b) shows $\|\mathbf{u}_{\text{ML}} - \mathbf{u}_{\text{obs}}\|$. Panel (c) shows the difference between the local CFG02 and uniform- C velocity errors; negative values indicate lower error for CFG02 and positive values indicate lower error for uniform C . Contours mark observed speeds of 100 (dotted), 500 (dashed), and 1000 m a^{-1} (solid). The cyan contour in panel (b) marks an inversion-reference velocity error of 100 m a^{-1} . Velocity fields are rendered from the original 450 m grid using display interpolation; all reported statistics use the original grid values. Both CFG02 and uniform C replace the inversion reference only inside the PIG holdout.

relative improvement in the fastest downstream flow. This spatial concentration explains how CFG02 can lower error across most of the catchment while retaining a large catchment-wide RMSE.

Locations where the inversion-reference error is at least 100 m a^{-1} comprise 3.9% of the evaluated PIG area but contain 47.1% of the CFG02 squared error. The remaining error is not confined to those locations: most locations with CFG02 error above 100 m a^{-1} lie outside regions where the inversion-reference error exceeds 100 m a^{-1} . Predictor support alone does not explain the spatial pattern. For CFG02, marginal support covers 97.5% of the evaluated PIG area, joint support covers 93.3%, and 91.9% satisfies both criteria. Of the net squared-error reduction relative to uniform C , 79.8% occurs within the area satisfying both support criteria. CFG02 nevertheless contains both successful and unsuccessful regions where predictor support is strong. CFG02 has a C RMSE of 0.85 across PIG. Locations with larger absolute differences from reference C also tend to have larger velocity errors, with a Spearman correlation of 0.67. This indicates a relatively strong positive association and is larger than the median within-square correlation of 0.32 for CFG02. The association between training-representation percentile and absolute C error is much weaker. C RMSE increases from 0.742 to 0.877 and 0.898 across the three categories of progressively sparser representation, but the Spearman correlation between representation percentile and absolute C error is only 0.031. This near-zero correlation indicates that having fewer nearby training examples does

not correspond to a consistent increase in local C error across PIG.

Interpreting the point-wise relationship

Taken together, the experiments separate three related questions: whether the predictor values in a held-out region are represented in the training data, whether the predicted C resembles the reference C , and whether the predicted C improves modeled velocity. The results show that these questions can have different answers. High predictor support can coexist with poor velocity performance, and sparser training representation does not consistently produce larger C errors. Agreement with the reference C is also related to, but does not determine, the resulting velocity error. The PIG experiment shows that the same distinctions remain important when the held-out region is expanded from a 50 km square to an entire glacier catchment.

The spatial differences in performance are not simply explained by the BedMachine diagnostics considered here. Reported bed-elevation uncertainty has only a weak relationship with local velocity error across the square tests (Appendix). Comparisons among BedMachine source classes are more limited because eight of the ten central squares contain only streamline-diffusion cells. Nevertheless, CFG02 velocity RMSE ranges from 3.5 to 136.3 m a⁻¹ across those eight squares, showing that differences in reconstruction class alone cannot explain the large spatial variation in performance. Reconstruction may still matter within a source class, for example if smoothing removes local bed variations that could help distinguish basal settings, but the present analysis does not isolate this effect.

More fundamentally, the available predictors may not capture all of the physical variability relevant to C . For example, Hoffman and others (2022) showed that small-scale basal roughness can strongly influence basal resistance, while such scales are not resolved in continental-scale datasets. The geophysical products also provide indirect information at relatively coarse spatial resolution. In addition, the inferred C depends on the adopted stress balance, basal law, and regularized inversion, rather than representing a unique physical property of the bed.

The point-wise formulation itself may also be limiting. The MLP uses only the predictor values at each location and cannot use information from neighboring locations. At the same time, membrane stresses couple the velocity response across space, so changes in C at one location can affect velocity elsewhere. This is consistent with theoretical limits on local basal inference, in which basal disturbances are filtered through spatially coupled ice flow before appearing at the surface (Gudmundsson, 2003; Raymond and

Gudmundsson, 2005; Gudmundsson and Raymond, 2008). The present experiments therefore identify several possible explanations for the remaining errors, but do not determine which mechanism is responsible in any individual region.

CONCLUSION

This study tested whether inversion-derived basal-friction control C can be represented by a point-wise relationship with commonly available surface, bed, and geophysical observables. We evaluated this relationship in ten spatially held-out regions across the Amundsen Sea sector and in a separate PIG catchment test by inserting the predicted C fields into Icepack and comparing the resulting velocities with observations.

The tested local observables contain information that can produce C fields that improve modeled velocity relative to uniform C in held-out regions, but that relationship is not spatially consistent across the Amundsen Sea sector. The ice-product configurations give the lowest median velocity errors, and the combined configurations improve on a spatially uniform- C reference in nine of ten squares. However, performance varies strongly among locations, adding predictors does not systematically improve performance, and poor performance can occur even where the held-out predictor values are well represented by the training data. Training representation provides no simple explanation either: sparser representation does not consistently correspond to larger C errors across the square tests. Withholding all of PIG shows the same limitation at catchment scale, with CFG02 reducing velocity RMSE by only about 8% relative to uniform C . Although CFG02 lowers local velocity error over most of PIG, the remaining error is strongly concentrated in fast flow.

The spatially withheld squares show that high predictor support can coexist with poor velocity performance. Agreement with the reference C is also only partly related to velocity performance. Local C error and velocity error are positively associated, but the relationship is not one-to-one, and closer agreement with C does not translate proportionally into better velocity performance. Under the framework considered here—the Amundsen Sea sector, the selected data products, a point-wise MLP, and a shallow-shelf/Weertman inversion—the tested surface, bed, and geophysical observables yield fitted controls that improve modeled velocity in most test regions. More broadly, the results show that such parameterizations should be tested in spatially held-out regions and evaluated through modeled velocity, rather than only through agreement with reference C . The same framework can be used to test other predictors and methods that use information from neighboring locations. The tested local observables do not support a

single point-wise relationship with inversion-derived C that is consistently successful across locations.

CODE AND DATA AVAILABILITY

The inversion, velocity-simulation, and model-training code is available at <https://github.com/racheetmatai/icepack-inverse>.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

STATEMENT ON THE USE OF AI

Generative AI tools were used to assist with language revision and consistency checking. All scientific decisions, analyses, interpretations, and final text were reviewed by the authors.

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ICE-FLOW EQUATIONS, GROUNDING TRANSITION, AND INVERSION METHOD

Icepack solves the depth-integrated shallow-shelf stress balance

$$\nabla \cdot (h\mathbf{M}) + \boldsymbol{\tau}_b = \rho_i g h \nabla s, \quad (7)$$

where h is ice thickness, s is surface elevation, ρ_i is ice density, g is gravitational acceleration, and $\boldsymbol{\tau}_b$ is the basal traction acting on the ice. For horizontal depth-averaged velocity $\mathbf{u}$, the membrane-stress tensor used here is

$$\mathbf{M} = 2\mu [\dot{\boldsymbol{\epsilon}} + \text{tr}(\dot{\boldsymbol{\epsilon}})\mathbf{I}], \quad \mu = \frac{1}{2}A^{-1/n}\dot{\epsilon}_e^{1/n-1}, \quad (8)$$

with $\dot{\boldsymbol{\epsilon}} = (\nabla\mathbf{u} + \nabla\mathbf{u}^T)/2$, $\dot{\epsilon}_e = [(\dot{\boldsymbol{\epsilon}} : \dot{\boldsymbol{\epsilon}} + \text{tr}(\dot{\boldsymbol{\epsilon}})^2 + \dot{\epsilon}_{\min}^2)/2]^{1/2}$, $\mathbf{I}$ the horizontal identity tensor, A the temperature-dependent rate factor, and $n = 3$ the Glen exponent. The modeled velocity is depth-averaged. Its comparison with the MEaSUREs surface velocity therefore uses the shallow-shelf assumption that vertical shear is negligible. We use the depth-averaged englacial temperature of Van Liefferinge and Pattyn (2013). This field is linearly interpolated to the model mesh and supplied in kelvin to Icepack's rate-factor function to calculate A (Shapiro and others, 2021). It is distinct from the surface-air-temperature predictor used by the MLP.

The basal law is Eqn 2, with the Icepack sliding exponent $n = 3$ in that equation. The dimensionless basal-friction control C is distinct from the dimensional basal-friction coefficient C_b that enters the traction law:

$$\begin{aligned} p_w &= \rho_w g \max(0, h - s), & p_i &= \rho_i g h, \\ \phi &= \begin{cases} 0, & p_i = 0, \\ \max[1 - (p_w/p_i)^p, 0], & p_i > 0, \end{cases} \\ C_b &= \phi C_0 \exp(C). \end{aligned} \quad (9)$$

Here ρ_w is seawater density, p_w is the hydrostatic water-pressure estimate, p_i is ice overburden pressure, $p = 1$ is the ramp exponent used in this study, and $C_0 = 0.01 \text{ MPa (m a}^{-1})^{-1/3}$. Consequently C_b has units $\text{MPa (m a}^{-1})^{-1/3}$, while C and ϕ are dimensionless. The ramp is continuous and piecewise, with kinks introduced by the maximum operations; it is not a differentiable smooth effective-pressure law.

Quadratic finite-element basis functions are used. Observed velocities are prescribed along the non-terminus boundaries, while an ocean-pressure traction condition is applied at the termini. In that terminus condition the water-depth magnitude is $d = \max(h - s, 0)$ and the normal-stress contribution is proportional to $(\rho_i g h^2 - \rho_w g d^2)/2$. Eleven Dirichlet velocity degrees of freedom without a colocated valid velocity pixel were filled from the nearest valid centered MEaSUREs pixel.

For each square, predicted C is applied only within the complete 130 km excluded footprint intersected

with the common eligible grounded area where $\phi > 0.1$. The PIG experiment uses the corresponding catchment mask. The uniform- C simulations use identical masks. Elsewhere, including through the grounding transition, the reference C is retained. This keeps the geometry, boundary conditions, and grounding treatment identical while holding the surrounding reference control fixed.

Sensitivity to the replacement region

We also compared the controlled design with an earlier calculation in which the predicted C , or the corresponding uniform- C value, replaced the inversion-reference control throughout the common eligible grounded sector. In that earlier design, the predicted control was therefore also used in the MLP training geography. In the controlled design used for the main results, replacement is restricted to the complete 130 km footprint for each square, or to the held-out regional geography, intersected with the same common eligible grounded mask. Predicted and uniform controls use identical masks, and both retain inversion-reference C outside the mask. The complete 40 km buffer around each central square is part of the replacement region.

The effects of correcting the observational reference and changing the replacement geography were evaluated separately. Replacing the mesh-interpolated observational values with the original paired MEaSUREs components changed the central-square MLP RMSE by a median of 0.07 m a^{-1} and a maximum of 0.95 m a^{-1} . Restricting replacement to the withheld geography then changed the central-square MLP RMSE by a median of 0.003 m a^{-1} and a maximum of 2.02 m a^{-1} . Neither change altered whether a configuration improved on uniform C in any square or which configuration had the lowest RMSE in a square. Differences were larger over the complete footprints. For CFG02 and CFG04, the median relative RMSE changed from 0.54 and 0.82 under sector-wide replacement to 0.64 and 0.86 under controlled replacement; both configurations improved on uniform C in nine of ten complete footprints in both designs. Across these 20 footprint comparisons, the median absolute change in relative RMSE was 0.04 and the largest was 0.28. The principal central-square findings are therefore robust to the two replacement designs, while the complete-footprint results are more sensitive to the surrounding control prescription.

The basal-friction control uses continuous quadratic finite elements. The replacement mask selects values at the finite-element control points, and outside-mask control-point values equal the inversion-reference C exactly. Within cells crossed by the geographic boundary, the quadratic basis connects inside- and outside-mask values. The transition can therefore be steep but is not a literal discontinuity in C ,

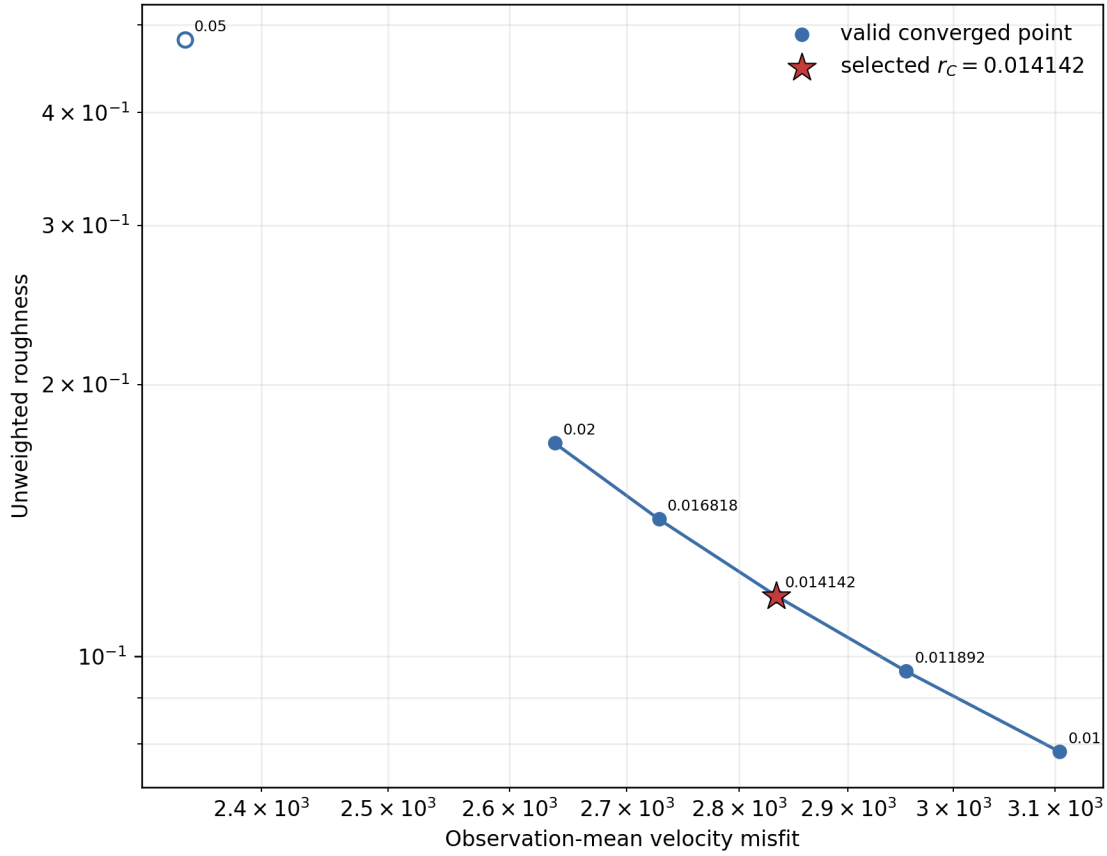


Fig. 8. L-curve for the sector-wide inversion. The maximum-curvature criterion selects $r_C \simeq 0.014$; smaller r_C corresponds to stronger smoothing in the objective function in Eqn 4. The neighboring candidate near $r_C \simeq 0.017$ has similar curvature, indicating a narrow elbow range. The open point at $r_C = 0.05$ has finite comparable objective values but reached the iteration limit and was not used in the selection.

and no explicit taper was applied. The central square is separated from the footprint boundary by the 40 km buffer. All controlled solves converged, but convergence and this separation do not show that boundary influence is negligible. The change in modeled velocity between the two designs is consistent with spatial coupling in the ice-flow model, but it does not isolate a uniquely non-local contribution because the surrounding control prescription and the transition at the replacement boundary change together.

The inversion was initialized with $C = 0$ and run for 300 optimization iterations. Figure 8 shows the L-curve used to select the regularization parameter. The maximum-curvature criterion selected $r_C \simeq 0.014$ from the refined candidates. The adjacent candidate, $r_C \simeq 0.017$, had curvature within approximately 4% of the selected value, indicating a narrow elbow range rather than a uniquely resolved regularization value.

SPATIAL SEPARATION AND BUFFER CHOICE

For each 50 km evaluation square, we exclude an additional 40 km buffer from MLP training and validation on all sides, giving a 130 km excluded footprint. The buffer is intended to reduce the influence of nearby, spatially correlated training data on the withheld-square test. We chose the 40 km distance before model fitting using the spatial scale of observed ice-speed variations.

Using MEaSUREs speed v , we formed

$$q(\mathbf{x}) = \ln[1 + v(\mathbf{x})/v_0], \quad v_0 = 1 \text{ m a}^{-1}. \quad (10)$$

The logarithm reduces the influence of the fastest outlets, while the offset keeps the transformation finite where speed approaches zero. We then removed the broad sector-scale variation using a normalized Gaussian average with a 50 km standard deviation, leaving local variations in log speed. These local variations have an estimated $1/e$ correlation length of 22.6 km, and their correlation falls to 0.108 at a separation of 40 km. Varying the Gaussian scale from 30 to 100 km gives $1/e$ correlation lengths of 16.1–46.0 km.

An independent estimate from fast-tributary widths gave a 90th-percentile width of approximately 36–38 km across speed thresholds of 100, 250, and 500 m a⁻¹. The 50 km test side therefore spans the coarsest 5 km predictors ten times and exceeds this tributary-width estimate. A 40 km buffer reduces the local log-speed-anomaly correlation to 0.108. Increasing the buffer to 50 km reduces it only to 0.072 while increasing each excluded footprint from 16,900 to 22,500 km², a 33% increase. Forty kilometres was therefore selected as a balance between separation from training rows and loss of training and validation data. The buffer separates the test square from nearby training locations at a distance where departures from the broad observed-speed trend have a much weaker spatial correlation than at nearby locations. The sector-wide inversion and non-local membrane-stress response still couple the withheld square to the surrounding ice.

ELIGIBLE ROWS, PREDICTOR CONSTRUCTION, AND SUPPORT

This section defines the common set of observation-grid rows used across all predictor configurations, describes how the derived predictors are constructed, and explains how predictor support is evaluated in the held-out regions.

We begin by defining the rows eligible for model training and evaluation. The MEaSUREs velocity raster

subset covering the study region contains 2,640,250 pixels. Requiring both observed velocity components to be finite and the MEaSUREs source flag to be positive retains 2,294,138 paired-velocity pixels; 1,622,598 of these lie inside the computational mesh. All twelve predictors are finite at those mesh-contained rows. Applying $\phi > 0.1$ leaves 1,530,992 common eligible rows, which are used by every predictor configuration. No additional rows are removed for individual configurations.

For the ten square experiments, excluding the 130 km footprint leaves 1,447,471–1,463,861 rows available for training and validation. Each MLP’s reproducible 90/10 split contains 1,302,723–1,317,474 training rows and 144,748–146,387 validation rows. The central-square evaluation sets contain 12,321 rows, except SQ03, which contains 12,432. For each training realization, every predictor configuration uses the same training locations and the same validation locations. Rows outside the common grounded eligible area retain inversion-reference C in every forward simulation. White or uncolored regions in maps denote locations outside the area shown for that experiment or outside that common mask; they were not removed on the basis of target values or model performance.

Surface elevation s , thickness h , and bed elevation b are represented as finite-element scalar fields. Their gradients are evaluated as finite-element vector expressions, interpolated component-wise to the observation grid, and summarized as $\|\nabla f\| = (f_x^2 + f_y^2)^{1/2}$ for $f \in \{s, h, b\}$. Driving-stress magnitude is

$$\tau_d = \rho_i g h \|\nabla s\|, \quad (11)$$

reported in the stress unit used by Icepack, MPa.

The alignment predictor is given by Eqn 5. The small-gradient scales were estimated by dividing representative elevation-error scales by the median mesh-edge length of 6553.4 m. For the surface, a 1 m elevation-error scale based on the reported REMA accuracy gives $\epsilon_s = 1.5 \times 10^{-4} \text{ m m}^{-1}$. For the bed, the median BedMachine bed-elevation error over grounded mesh points is 35.31 m, giving $\epsilon_b = 5.4 \times 10^{-3} \text{ m m}^{-1}$. The resulting values were rounded and fixed before model evaluation. The alignment value is clipped to $[-1, 1]$. The ϵ_b and ϵ_s terms in the denominator reduce the magnitude of the alignment toward zero where either gradient is small relative to its corresponding scale.

The selected ice-product and geophysical-product subsets came from an earlier catchment-level comparison of modeled velocity and were specified before the present spatial results were examined. Table 4 summarizes the native spatial resolution and interpretation of the products used. Interpolation to the observation grid increases the number of sampled values but does not add new spatial information.

Table 4. Sources and source-grid spacing of the predictors. Nominal cells in a 50 km square are geometric counts before masks and are not necessarily independent samples. Derived gradients retain the resolution limits of their source fields.

Predictor(s)	Product/source	Source grid	Nominal cells	Interpretation in this study
s, h, b and their gradients	BedMachine Antarctica v2	500 m	$\sim 10,000$	Geometry products. Fast-flow bed reconstruction uses mass conservation and velocity, and is not a direct measurement of local basal processes.
τ_d	Derived from BedMachine h, s	500 m source fields	$\sim 10,000$	Large-scale driving-stress estimate derived from thickness and surface slope.
T_s	Quantarctica surface- air-temperature raster	5 km	~ 100	Large-scale surface thermal boundary, not a direct observation of basal temperature.
Q_b	Quantarctica geothermal-heat-flux raster	5 km	~ 100	Model-derived regional basal heat-flux estimate; interpolation does not add fine-scale information.
m_a	ADMAP-2S EPSG:3031 grid	5 km	~ 100	Magnetic anomaly containing broad information about crustal magnetization; it does not uniquely identify basal material.
g_d	AntGG2021 oriented EPSG:3031 grid	5 km	~ 100	Gravity disturbance containing information about subsurface density; it does not uniquely identify rock type or subglacial water.
a_{bs}	Derived from BedMachine b, s	500 m source fields	$\sim 10,000$	Relative alignment of two geometry gradients; does not use flow direction.

Marginal support is evaluated separately for each predictor. A held-out point satisfies the marginal criterion when every predictor included in that configuration lies within the 1st–99th percentile range of the corresponding eligible non-held-out rows.

Joint support evaluates the full combination of predictor values. Predictor values on a fixed 5 km grid covering the sector are first converted to rank-based normal scores by matching their ranks to standard-normal quantiles (Beasley and others, 2009). We then use principal component analysis (PCA; Jolliffe and Cadima, 2016) to reduce the predictor space while retaining at least 99% of the variance of the transformed predictors, and scale the retained components to unit variance. The same transformation is used for all spatial experiments.

To define the joint-support criterion used in square selection and later evaluation, we use this grid to establish what counts as a sufficiently similar predictor combination. For each grid point, we calculate the distance in transformed predictor space to the most similar point located at least 40 km away geographically. The 95th percentile of these distances defines the joint-support threshold for that predictor configuration. For each held-out point, we then find the most similar 5 km reference-grid point within the corresponding training region. The held-out point satisfies the joint criterion when this distance does not exceed the threshold.

Reported predictor support is the area fraction satisfying both the marginal and joint criteria. All held-out points are retained in the modeled velocity evaluation regardless of support.

BASAL-FRICTION CONTROL: DISTRIBUTIONS AND COMPARISONS

We first check whether the reference C values in each held-out region fall within the range of values available in the training-and-validation region. This is separate from predictor support, which considers only the input variables. Reference C was not used to select the square locations, define predictor support, or remove evaluation rows.

Figures 9, 10 and Table 5 compare the reference C distributions in the training-and-validation and held-out regions. All predictor configurations within a given experiment use the same combined training-and-validation and held-out rows, so the reference- C distributions are also the same for all configurations in that experiment.

These comparisons show how the reference- C range in each held-out region overlaps that in the training-and-validation region; modeled velocity tests whether the predicted control reproduces the observed flow.

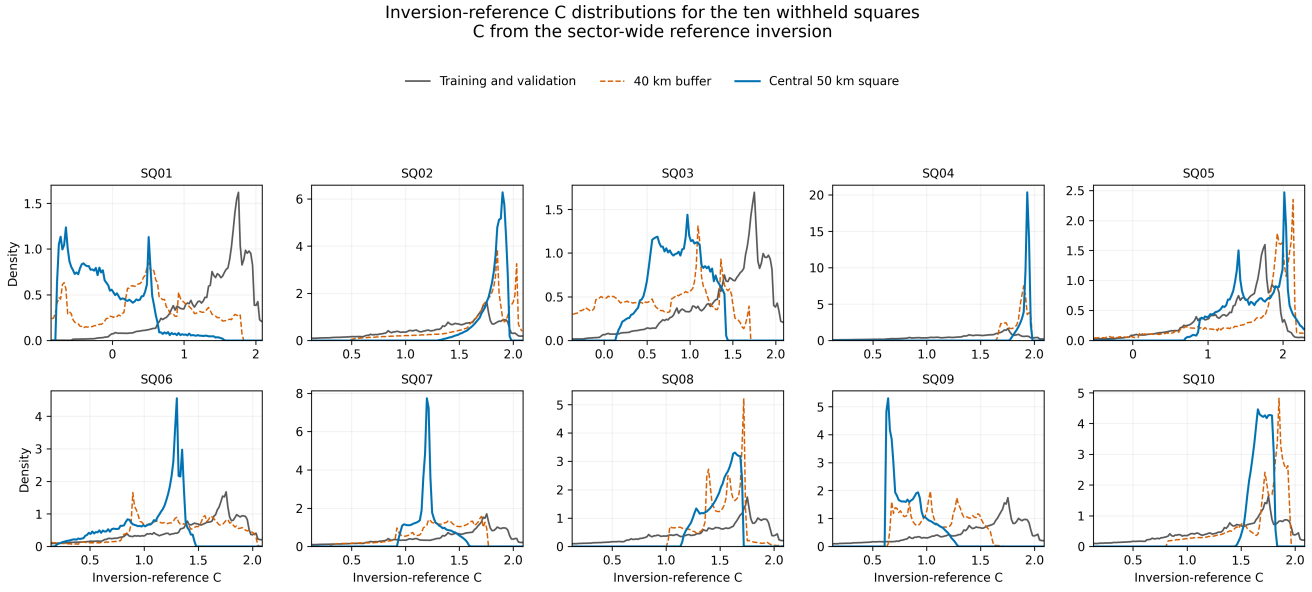


Fig. 9. Reference- C distributions for the ten withheld squares. Each panel compares the training and validation rows with the central 50 km evaluation square and the surrounding buffer. The horizontal range is chosen separately for each panel to show the central part of the distributions clearly. Each density is normalized using all rows in its set, including values outside the displayed range. Numerical 1st–99th-percentile intervals and the percentage of held-out values within the training-and-validation interval are reported in Table 5. Reference C was not used to select square locations or remove evaluation rows.

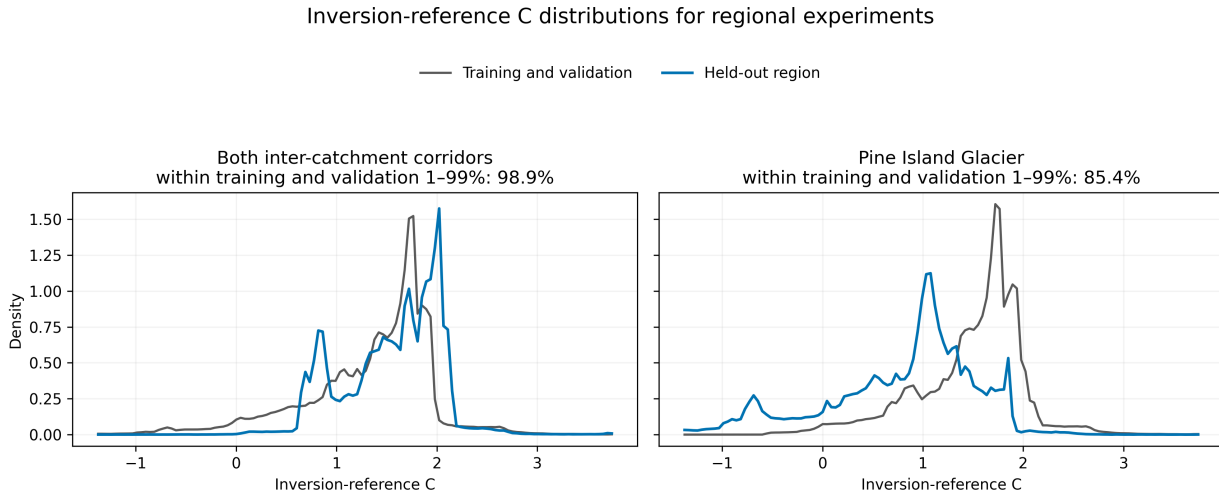


Fig. 10. Reference- C distributions for the two regional tests. Each panel compares the training and validation rows with the complete held-out region. The percentage shown is the fraction of held-out C values that falls within the 1st–99th-percentile range of the training and validation data.

Table 5. Reference- C distributions in the training-and-validation and held-out regions. Intervals are the 1st–99th percentiles of dimensionless C . The final column gives the percentage of held-out rows whose C values fall within the corresponding training-and-validation interval.

Experiment	Training and validation rows	Held-out rows	Training and validation interval	Held-out interval	Held-out within training range (%)
SQ01	1,448,295	12,321	−0.22–2.67	−0.76–1.30	51.9
SQ02	1,447,471	12,321	−0.69–2.67	1.43–1.96	100.0
SQ03	1,450,403	12,432	−0.69–2.67	0.21–1.40	100.0
SQ04	1,447,471	12,321	−0.69–2.67	1.80–1.96	100.0
SQ05	1,459,429	12,321	−0.65–2.66	0.84–2.35	100.0
SQ06	1,447,471	12,321	−0.69–2.67	0.28–1.43	100.0
SQ07	1,447,471	12,321	−0.69–2.67	0.95–1.55	100.0
SQ08	1,453,825	12,321	−0.69–2.66	1.18–1.71	100.0
SQ09	1,458,393	12,321	−0.69–2.66	0.62–1.23	100.0
SQ10	1,463,861	12,321	−0.69–2.66	1.50–1.81	100.0
Inter-catchment	1,241,683	289,309	−0.72–2.65	0.49–2.66	98.9
PIG	1,303,307	227,685	−0.08–2.69	−1.28–2.05	85.4

The reference C is one regularized, model-dependent inversion solution, and modeled velocity remains the primary evaluation in this study.

We measure agreement with the inversion-reference control using C RMSE and the coefficient of determination R_C^2 . The latter is one minus the sum of squared prediction errors divided by the sum of squared deviations of reference C from its mean over the same evaluated locations.

Table 6 reports agreement with the inversion-reference control on the validation rows. Each MLP was evaluated on its own validation rows, so the table reports the median of the ten member-level results for each square-configuration experiment. Validation C RMSE is lower and R_C^2 higher for the other five configurations than for CFG04. The contrast with the spatially held-out results in Table 7 shows that agreement on randomly withheld rows within the training region does not imply comparable transfer to a spatially separated square.

We also compare the predicted C fields directly with the reference C to measure how closely the MLP reproduces the inversion result. Table 7 reports C RMSE and R_C^2 for every central square. Negative R_C^2 values are retained; a negative value means that the predicted C has larger squared error than using the mean reference C within that square. The median R_C^2 values across the ten squares are −1.39, −1.35, −7.97, −6.52, −1.41, and −1.21 for CFG01–CFG06, respectively. Poor agreement with the reference C

Table 6. Validation agreement with the inversion-reference control. Each entry is the median C RMSE / R_C^2 across the ten MLP members for that square-configuration experiment, calculated on each member’s own validation rows. RMSE is in the dimensionless control C , and R_C^2 is dimensionless. The validation rows lie outside the excluded 130 km footprint and were used for selection of the fitted MLP; they are not independent test data.

Square	CFG01	CFG02	CFG03	CFG04	CFG05	CFG06
SQ01	0.077/0.984	0.075/0.985	0.056/0.992	0.363/0.652	0.020/0.999	0.022/0.999
SQ02	0.078/0.987	0.081/0.986	0.057/0.993	0.387/0.680	0.023/0.999	0.022/0.999
SQ03	0.072/0.988	0.077/0.986	0.070/0.989	0.383/0.661	0.022/0.999	0.021/0.999
SQ04	0.080/0.986	0.079/0.986	0.056/0.993	0.393/0.666	0.023/0.999	0.022/0.999
SQ05	0.076/0.987	0.077/0.986	0.055/0.993	0.364/0.697	0.023/0.999	0.021/0.999
SQ06	0.073/0.988	0.080/0.986	0.057/0.993	0.382/0.681	0.022/0.999	0.022/0.999
SQ07	0.077/0.987	0.077/0.987	0.058/0.993	0.388/0.681	0.021/0.999	0.022/0.999
SQ08	0.077/0.988	0.079/0.987	0.056/0.993	0.397/0.669	0.022/0.999	0.023/0.999
SQ09	0.077/0.987	0.079/0.986	0.057/0.993	0.384/0.676	0.022/0.999	0.022/0.999
SQ10	0.077/0.987	0.080/0.986	0.058/0.993	0.391/0.676	0.023/0.999	0.022/0.999

can still coexist with improved modeled velocity, so these statistics answer a different question from the velocity comparison.

For the PIG holdout experiment, the median validation C RMSE / R_C^2 across the ten MLP members is 0.081/0.981 for CFG01, 0.077/0.983 for CFG02, and 0.056/0.991 for CFG03. These values were calculated on each member’s own validation rows outside the withheld PIG region. Despite this strong validation agreement, the corresponding median predicted fields have PIG holdout C RMSE / R_C^2 values of 0.92/−0.49, 0.85/−0.28, and 1.23/−1.69, respectively. Thus, performance on randomly withheld validation rows does not predict spatial transfer to the withheld catchment.

The inter-catchment experiment provides an additional regional test by withholding the complete PIG–Thwaites and Thwaites–Dotson corridors. This tests transfer from the named glacier catchments to the regions between them. For the two corridors together, the holdout C RMSE / R_C^2 values are 0.81/−1.86, 0.33/0.53, and 0.33/0.52 for CFG04–CFG06. Across both corridors, the median predicted C fields give velocity RMSE values of 470.3, 119.0, and 109.3 m a^{−1} for CFG04, CFG05, and CFG06, respectively, compared with 578.4 m a^{−1} for uniform C and 28.1 m a^{−1} for the inversion reference. The corresponding relative RMSE values are 0.81, 0.21, and 0.19. The two corridors are spatially separate, and the intervening region is not included in the calculation. All three configurations reduce velocity RMSE relative to uniform C over the combined corridor area, with lower errors for CFG05 and CFG06 than for CFG04. For these

Table 7. Comparison between predicted and reference C in the ten central squares. Each entry gives C RMSE / R_C^2 for the median predicted C field. RMSE is in dimensionless C units.

Square	CFG01	CFG02	CFG03	CFG04	CFG05	CFG06
SQ01	0.42/0.25	0.41/0.29	1.44/−8.00	1.21/−5.42	0.58/−0.47	0.52/−0.20
SQ02	0.22/−2.69	0.22/−2.46	0.88/−56.51	0.61/−26.00	0.17/−1.04	0.15/−0.69
SQ03	0.38/−0.63	0.37/−0.61	0.54/−2.31	0.54/−2.32	0.35/−0.38	0.33/−0.29
SQ04	0.20/−29.81	0.20/−30.02	0.36/−98.61	0.33/−80.67	0.16/−18.76	0.16/−18.71
SQ05	0.37/0.03	0.37/0.02	0.46/−0.50	0.48/−0.63	0.41/−0.21	0.42/−0.25
SQ06	0.52/−1.83	0.51/−1.72	0.51/−1.77	0.47/−1.35	0.33/−0.14	0.34/−0.19
SQ07	0.24/−2.44	0.23/−2.18	0.57/−17.72	0.52/−14.99	0.30/−4.13	0.31/−4.77
SQ08	0.20/−0.96	0.20/−0.98	0.28/−2.78	0.31/−3.43	0.24/−1.77	0.24/−1.73
SQ09	0.74/−18.37	0.74/−18.49	0.87/−26.27	0.49/−7.63	0.53/−8.90	0.51/−8.38
SQ10	0.10/−0.79	0.10/−0.83	0.23/−7.93	0.31/−14.95	0.31/−15.21	0.27/−11.59

velocity calculations, all four controls replace the inversion reference only inside the combined corridor geography and retain the same reference C outside it, matching the controlled comparison used for the square and PIG CFG02 results.

Finally, we examine the small number of unusually large reference- C values because mean-square loss gives large prediction residuals greater weight. Across all eligible rows, C has a median of 1.54, a 1st–99th-percentile range of -0.68 to 2.65 , a 99.9th percentile of 3.76 , and a maximum of 24.18 . Values greater than 5 occur in only 821 of 1,530,992 rows (0.054%). They are more common near the grounding transition but are not confined to it: 261 occur where $0.1 < \phi \leq 0.2$. The scaling used for training changes the numerical scale of C but does not selectively reduce the contribution of large prediction residuals. The largest 0.1% of squared deviations from the target median account for 12.5% of the total squared deviation.

For the MLPs selected by minimum validation MSE, rows with $C > 5$ account for a median of 1.59% of validation squared error; excluding them from validation scoring reduces median validation RMSE by 0.77% and does not materially alter the configuration comparisons. No such rows occur in the ten central squares or the PIG evaluation region, so we retain them rather than introduce a target-based exclusion and retraining step. These rare large values are features of the inversion-derived target and should not be interpreted as direct evidence for extreme physical basal resistance.

Training representation and prediction error

We also test whether prediction errors are related to how densely the predictor values at each held-out location are represented in the training data. For CFG02 and CFG04 in the ten central 50 km squares, and CFG02 in PIG, we calculate d_{20} for each held-out row, defined as the distance to its 20th-nearest training row. Nearest-neighbor distances underlie multivariate density estimation (Loftsgaarden and Quesenberry, 1965); here d_{20} describes training representation rather than estimating a probability density. Distances are calculated after applying the same transformation used for joint support: predictor ranks are mapped to normal scores, followed by PCA retaining at least 99% of the variance, with the retained components scaled to unit variance. This transformation is fitted to the eligible sector-wide 5 km grid. Smaller d_{20} values indicate that many similar predictor combinations occur in the training data, whereas larger values indicate sparser representation. Unlike the joint-support calculation, no minimum geographic separation is imposed here; all training rows are used, including nearby rows and repeated predictor combinations.

For each MLP realization, 10,000 training rows are sampled uniformly without replacement. The d_{20} value for each sampled training row is calculated against the complete training set, excluding only that row itself. These 10,000 values provide a reference distribution for the held-out rows. Each held-out d_{20} value is assigned the percentile

$$p = 100 \frac{N(d_{20,\text{train}} < d_{20,\text{test}}) + \frac{1}{2}N(d_{20,\text{train}} = d_{20,\text{test}})}{10,000}. \quad (12)$$

For each held-out row, we use the median percentile across the ten MLP realizations for that experiment and configuration. Higher percentiles indicate sparser representation in the training data. We group the held-out rows into three categories: $p \leq 50$, $50 < p \leq 95$, and $p > 95$. The neighbor count and category boundaries were fixed before examining prediction errors.

We examine the original training data without rebalancing. The predictors are continuous and several are physically or mathematically related. Balancing each predictor separately would not necessarily balance combinations of predictors and could change the relative frequency of the combinations present in the original data. Instead, this analysis measures how densely the existing training data represent each held-out predictor combination.

To compare training representation with prediction error, the median predicted C field is evaluated on the same 450 m observation-grid rows used for the representation analysis. The median is first calculated

on the model mesh and then interpolated to the observation grid in the same way as the reference C . All central-square rows are retained. The PIG analysis uses the same procedure over its complete eligible held-out area. Within each representation category, we calculate C RMSE, mean signed difference, and mean absolute difference from the reference C . Velocity RMSE and relative RMSE are calculated over the same rows.

The whole-square C RMSE values calculated on the regular 450 m grid differ by at most 2.37% from the area-weighted values in Table 7, which are calculated on the finite-element mesh. All C errors are reported in the original dimensionless C units.

Table 8 tests whether C prediction error increases as training representation becomes sparser. For CFG02, C RMSE increases from the better-represented to the more sparsely represented categories in seven of the ten squares. However, this pattern is not consistent across the full set of experiments. The median C RMSE across the ten squares is 0.299, 0.297, and 0.301 for $p \leq 50$, $50 < p \leq 95$, and $p > 95$, respectively, and the median within-square Spearman correlation between representation percentile and absolute C error is only 0.069. Thus, although some individual squares show larger C errors where similar training examples are sparser, training representation does not provide a general explanation for C prediction error. CFG04 shows even less evidence of such a relationship: only one square has increasing C RMSE across the three categories, and the median within-square correlation is -0.052 .

To test whether the weak relationship between training representation and C error is being obscured by differences in ice speed, we repeat the analysis within observed-speed classes. For each class, the correlation is between representation percentile and absolute C error. Below 100 m a^{-1} , the median within-square correlations are 0.080 for CFG02 and -0.113 for CFG04. Between 100 and 500 m a^{-1} , they are 0.079 and 0.015, respectively. These values remain close to zero, indicating that sparser training representation is not consistently associated with larger C error within either the slow- or moderate-flow classes. Only one square contributes to the $500\text{--}1000 \text{ m a}^{-1}$ class, so there are not enough repeated spatial tests to draw the same conclusion for faster flow.

Relative velocity RMSE also does not vary consistently with training representation. In SQ06, CFG02 performs worse than uniform C in the best-represented category (relative velocity RMSE 1.15) but improves on uniform C in the sparsest category (0.96). In SQ05, C RMSE decreases from 0.416 to 0.337 between these categories (Table 8), while relative velocity RMSE increases from 0.37 to 0.81. In SQ02–CFG04, the best-represented category has a C RMSE of 0.608 and a relative velocity RMSE of 1.45, whereas the

Table 8. Prediction error in C relative to the inversion reference and training representation in the central squares. The three columns give dimensionless C RMSE for increasingly sparse representation categories. Spearman ρ relates the representation percentile of each row to its absolute C error within each square. The last row for each configuration gives the median across the ten squares.

Configuration	Square	$p \leq 50$	$50 < p \leq 95$	$p > 95$	ρ
CFG02	SQ01	0.371	0.410	0.418	0.076
	SQ02	0.206	0.232	0.265	0.107
	SQ03	0.370	0.384	0.411	0.086
	SQ04	0.201	0.204	0.211	0.056
	SQ05	0.416	0.351	0.337	-0.141
	SQ06	0.493	0.516	0.543	0.035
	SQ07	0.227	0.242	0.245	0.090
	SQ08	0.175	0.212	0.198	0.133
	SQ09	0.731	0.735	0.753	0.062
	SQ10	0.130	0.103	0.110	-0.087
	Median	0.299	0.297	0.301	0.069
CFG04	SQ01	1.100	1.298	1.058	-0.101
	SQ02	0.608	0.634	0.470	-0.049
	SQ03	0.512	0.572	0.587	0.139
	SQ04	0.375	0.364	0.236	-0.637
	SQ05	0.473	0.508	0.422	0.065
	SQ06	0.383	0.515	0.474	0.224
	SQ07	0.634	0.477	0.466	-0.257
	SQ08	0.348	0.307	0.309	-0.044
	SQ09	0.517	0.491	0.411	-0.055
	SQ10	0.299	0.317	0.302	-0.094
	Median	0.492	0.499	0.444	-0.052

sparsest category has lower C RMSE (0.470) and relative velocity RMSE (1.34). These examples show that sparser training representation does not consistently correspond to larger relative velocity RMSE. They also show that closer agreement with the reference C does not necessarily produce lower relative velocity RMSE.

MLP TRAINING AND STATISTICAL COMPARISONS

This section describes the MLP architecture and training procedure, the repeated training realizations used for each experiment, and the statistical comparisons used to summarize differences among configurations.

The MLP has ten hidden layers of 200 units. Each hidden layer is followed by batch normalization and a SiLU activation, and the final layer is linear. Predictors and C are robustly scaled using medians and interquartile ranges fitted on training rows only. Adam uses an initial learning rate of 10^{-3} and batch size 1024, with reproducible epoch-wise shuffling. The learning rate is multiplied by 0.2, down to 10^{-6} , after 15 epochs without a validation-MSE decrease of at least 10^{-4} . Training stops after 45 epochs without improvement in validation MSE, and the fitted MLP with the lowest unpenalized validation MSE is used.

A separate CFG06 calculation selected the L2 penalty without using any of the ten central squares. Its fixed 70/20/10 split contained 1,407,671 rows: 985,369 for training, 281,534 for validation, and 140,768 for testing. L2 penalties of 0, 10^{-6} , 10^{-5} , and 10^{-4} were applied to the hidden-layer weights and compared using unpenalized validation MSE. The respective minimum values were 0.001207, 0.001332, 0.001583, and 0.003117. No nonzero penalty reduced validation MSE below that of the zero-penalty model, so all final MLP trainings used $\lambda_{L2} = 0$. The test rows were not used to choose the penalty.

We did not optimize the network architecture separately for each predictor configuration or held-out region. Instead, we used the same relatively large network, with ten hidden layers of 200 units, for all experiments. This architecture provides substantial flexibility to represent nonlinear relationships between the predictors and C , while keeping the network design fixed across configurations and spatial tests. Overfitting is limited through a separate validation set, learning-rate reduction, early stopping, and restoration of the model with the lowest validation MSE. The training and validation histories shown below provide an additional check, with no systematic separation between them during training. A full architecture search would require separate tuning across multiple predictor configurations and spatial splits and was beyond the scope of this study. The chosen architecture should therefore be viewed as a fixed, flexible modeling choice rather than an optimized architecture.

Each experiment–configuration pair uses ten reproducible training realizations with different training/validation row assignments, initial weights, and epoch shuffles. We trained 660 MLPs in total. The regional tests use CFG01–CFG03 for PIG and CFG04–CFG06 for the inter-catchment corridors. For each experiment–configuration pair, the ten predicted C fields are combined by taking the median at each mesh vertex, giving 66 median fields. The controlled forward evaluation in the main text uses the 60 square fields and the PIG CFG02 field. The individual realizations are used to assess sensitivity to the training procedure and to construct the median prediction; they are not treated as independent spatial tests or as estimates of predictive uncertainty.

For the velocity-error calculations, modeled velocity is interpolated to the pixel centres of the regular 450 m MEaSUREs observation grid and compared with the original paired raster components there. Squared vector residuals are averaged over the retained grid points before taking the square root. Because the observation grid has uniform spacing, this provides an equal-area approximation to the spatial mean rather than an exact finite-element integration.

For the spatially uniform- C reference, the mean inversion-reference C is calculated from finite-element cells in the non-held-out region. Cell means of C are weighted by cell area. Geographic membership and the criterion $\phi > 0.1$ are evaluated at cell centers, with positive thickness and finite C also required. The resulting value therefore uses model cells rather than observation-grid row means. It is applied over the same withheld footprint or catchment used for the predicted C field, with inversion-reference C retained elsewhere.

The ten squares are treated as the independent spatial units for statistical comparison. The PIG and inter-catchment experiments are secondary regional tests and do not add independent units to these comparisons. The spatially uniform- C simulation tests whether predictor-dependent spatial variation reduces velocity RMSE relative to uniform C ; it is not a comparison with another learning algorithm. Results are reported for every square and summarized using the median and interquartile range across the ten squares. Exact paired sign tests compare predictor configurations chosen in advance. The individual MLP realizations, and the central square and full footprint from the same experiment, are not counted as additional independent units. All 66 median predicted C fields are finite. Predictions were not clipped to the range of the inversion-reference C .

The complete histories in Fig. 11 show that the training and validation losses decrease and then approach a plateau. Median validation MSE decreases from its initial value by factors of 19.5, 19.4, 37.7, 1.68,

148, and 162 for CFG01–CFG06, respectively. Training and validation histories remain close throughout training, with no systematic growth in the gap between them. This provides no clear evidence that the relatively large network is increasingly fitting the training rows at the expense of the validation rows. Validation agreement with the reference C is high for CFG01, CFG02, CFG03, CFG05, and CFG06, while CFG04 shows a much smaller reduction in validation loss and higher validation C RMSE and lower R_C^2 . However, CFG04 still reduces velocity RMSE relative to uniform C in seven of the ten spatially withheld squares. The main implication is that successful fitting of C on random validation rows within the training region does not by itself predict performance in a spatially separated region. The common architecture and training procedure provide a controlled comparison among predictor configurations, while the spatially withheld tests assess whether the learned relationships produce useful modeled velocities elsewhere.

We used paired sign tests to assess whether one configuration consistently had lower velocity RMSE than another across the ten withheld squares (Hollander and others, 2013, Chapter 3, Section 3.4). All 15 possible pairwise configuration comparisons were tested; none was statistically significant after Holm correction for multiple comparisons (Holm, 1979). All 15 Holm-adjusted p -values are 1.0. For the five comparisons corresponding directly to predictor additions or removals in the configuration design, the unadjusted two-sided p -values were 1.0 for CFG02–CFG01, 1.0 for CFG04–CFG03, 0.344 for CFG05–CFG02, 0.344 for CFG05–CFG04, and 1.0 for CFG06–CFG05. Thus, although some configurations have lower median RMSE or perform better in more squares, no configuration shows a statistically significant consistent advantage over another across the ten spatial tests.

BedMachine reconstruction and velocity error

Because several predictors are derived from BedMachine, we also examined whether differences in modeled velocity performance could be related to the BedMachine reconstruction itself. We considered two quantities: the source class, which identifies the method used to construct the bed elevation, and the reported bed-elevation error. Neither quantity was used as an MLP predictor or to select or exclude evaluation rows. The analysis uses the ten central 50 km squares and the velocity from each configuration’s median- C forward simulation.

For each square and configuration, we calculated the Spearman correlation between reported bed-elevation error and local velocity-vector error, $\|\mathbf{u}_{\text{ML}} - \mathbf{u}_{\text{obs}}\|$. Table 9 summarizes the ten within-square correlations. Their small magnitudes and changing signs show that larger reported bed-elevation errors

Training histories for the ten withheld squares (100 MLPs per configuration)

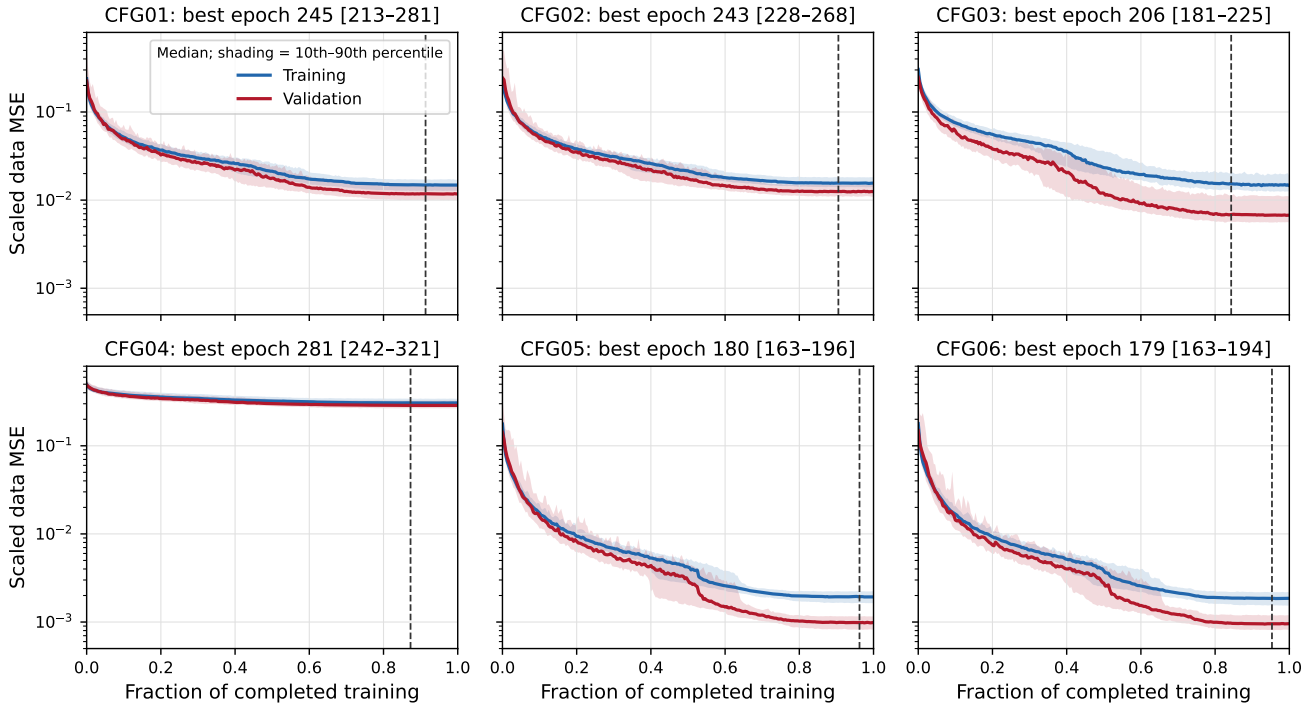


Fig. 11. Training histories for all 600 primary-square MLP runs, comprising ten squares and ten MLPs for each configuration. To retain every run in the summary, each history is scaled from its first epoch (training fraction zero) to the end of training (training fraction one). Lines show median training and validation data MSE, and shading spans the 10th–90th percentiles across the 100 runs in each configuration. The vertical dashed line marks the median position of the fitted MLP with the lowest validation MSE; panel titles give the median and interquartile range of its absolute epoch. MSE is computed after robust scaling fitted only on the training rows and is used to fit the MLP, not to measure held-out modeled velocity performance. For each experiment, its central square was not used to select a fitted MLP.

Table 9. Spearman correlations between reported BedMachine bed-elevation error and local velocity error in the ten central squares. Each configuration is summarized by the median and range of its ten within-square correlations.

Configuration	Median	Minimum	Maximum
CFG01	-0.005	-0.117	0.139
CFG02	-0.018	-0.135	0.137
CFG03	0.012	-0.151	0.082
CFG04	0.041	-0.139	0.138
CFG05	-0.062	-0.218	0.193
CFG06	-0.059	-0.214	0.181

did not consistently coincide with larger velocity errors.

BedMachine source class records how the geometry supplied to the MLP was reconstructed. Class 5 denotes streamline-diffusion interpolation, which smooths thickness preferentially along flow; it is not a direct bed measurement. Eight central squares contain only class 5 rows. SQ05 contains 838 mass-conservation rows (class 2), 174 interpolation rows (class 3), and 11,309 class 5 rows; SQ06 contains 12, 18, and 12,291 rows in these classes, respectively. These are counts of evaluated observation-grid rows.

Because reconstruction method changes little within most squares, these tests cannot separate its effect from other differences between locations. Performance also varies among squares using the same method, so differences between source classes cannot alone account for the observed spatial variation. We did not test source class as an additional predictor or quantify how much along-flow smoothing changes the predictor gradients and modeled velocity.